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MacroRisk-agent: a retrieval-augmented multi-agent framework with deterministic validation gates for auditable stress testing

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Abstract

Large language model (LLM)-based multi-agent systems face three open challenges in high-stakes decision support: hallucination control in structured output generation, coupling of semantic reasoning with deterministic numerical simulation, and end-to-end auditability of multi-step agentic workflows. We present *MacroRisk-Agent*, a multi-agent architecture that addresses these challenges in the domain of financial stress testing—a setting that demands strict output validity, reproducibility, and regulatory traceability. MacroRisk-Agent introduces three technical contributions: (1) a retrieval-augmented generation (RAG) pipeline with deterministic validation gates that converts natural-language stress intent into executable, schema-constrained parameter objects, achieving 92.6% validity rate versus 78.4% for RAG-augmented baselines ($\mathbf{p < 0.001}$); (2) a constraint-coupled multi-agent simulation that models endogenous amplification through fire-sale, margin, and liquidity feedback loops, producing a 52% higher systemic-loss estimate than the one-pass configuration under the calibrated experimental setting; and (3) a counterfactual re-simulation framework that validates proposed interventions under identical scenario constraints, achieving 68.6% simulation-defined top-1 decision success rate. We evaluate MacroRisk-Agent as a *proof-of-concept* on public financial datasets (FRED, Yahoo Finance), a library of 150 curated crisis events, and 10 synthetic institutional portfolios; reported improvements are statistically significant across scenario validity, propagation modeling, and intervention quality *within this synthetic evaluation environment*. We position the framework as a proposed architecture—rather than an empirically validated, deployment-ready system—for building hallucination-controlled, provenance-tracked LLM-agent pipelines that are designed to improve auditability in high-stakes settings; validation on real institutional data is left to future work.

Introduction

The deployment of large language models (LLMs) in structured decision-making requires solving three interlocking technical problems that remain open: how to prevent hallucination in parameterized output generation, how to couple language-model reasoning with deterministic simulation engines, and how to maintain end-to-end provenance in multi-step agentic workflows[1,2]. These challenges are particularly acute in high-stakes domains—such

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as financial risk assessment, clinical decision support, and infrastructure planning—where generated outputs must satisfy hard constraints, undergo formal validation, and produce audit trails that withstand external scrutiny.

Technical challenges

We identify five core technical gaps (GAP-1 through GAP-5) that arise when applying LLM-agent systems to structured decision support, instantiated here through the lens of financial stress testing:

1. **Constrained Generation (GAP-1).** LLMs generating structured outputs (e.g., parameterized scenario specifications) lack built-in validity guarantees. Without schema enforcement and domain-specific validation, generated outputs frequently violate feasibility constraints, contain numerically implausible values, or hallucinate parameters unsupported by retrieved evidence[3,4].
2. **Semantic-to-Numeric Coupling (GAP-2).** Translating language-level specifications (natural-language narratives, qualitative descriptions) into numerical simulation parameters requires a principled interface between semantic reasoning and deterministic computation. Existing approaches either delegate numerical tasks to LLMs (introducing hallucination risk) or treat the two modalities in isolation[5,6].
3. **Feedback Modeling (GAP-3).** Standard LLM inference is single-pass: it produces outputs without modeling the dynamic, multi-step state evolution that arises when heterogeneous agents interact under constraints. Capturing emergent feedback loops—where one agent's action changes the environment for others—requires coupling LLM-generated specifications with iterative multi-agent simulation[7,8].
4. **Multi-level Aggregation (GAP-4).** Many decision-support tasks require simultaneous computation of micro-level (per-entity) and macro-level (system-wide) outcomes within a unified simulation. Siloed computation prevents identification of emergent systemic risks that arise from the interaction of individually rational behaviors[9,10].
5. **Auditability (GAP-5).** In regulated and high-stakes settings, every generated output must be linked to retrievable evidence for post-hoc review. Existing LLM-agent frameworks provide limited support for end-to-end provenance tracking that connects generated artifacts to their upstream evidence, model versions, and intermediate reasoning steps[11].

To address these challenges, we propose **MacroRisk-Agent**, a multi-agent architecture that combines retrieval-augmented generation with deterministic simulation engines and provenance-tracked reporting. We instantiate MacroRisk-Agent in the domain of financial stress testing—a setting that imposes strict requirements on output validity, reproducibility, and traceability—but the architectural principles apply broadly to any domain requiring hallucination-controlled, auditable LLM-agent systems. As illustrated in Fig. 1, the framework addresses GAP-1 through GAP-5 by: (i) using a Scenario Agent with RAG and multi-stage validation gates (Definitions 1–3) to produce schema-constrained, evidence-grounded scenario objects; (ii) simulating shock propagation through interacting agents with endogenous feedback under funding and margin constraints; (iii) consolidating entity-level and system-level metrics in a unified analytics layer; and (iv) closing the governance loop through a counterfactual re-simulation framework that validates interventions and produces provenance-tracked reports.

Contributions

The central contribution of this paper is an *architectural* one—a validation-gate-and-provenance design that makes a multi-agent LLM pipeline auditable—which we instantiate and stress-test in financial stress testing rather than offering as a new financial-risk or systemic-risk model. Stated as a single primary contribution: *a constrained LLM-to-simulation interface that converts evidence-grounded stress narratives into validated, executable scenario specifications, coupled to traceable counterfactual simulation.* The remaining components—the multi-agent amplification engine, the systemic-risk analytics, and the intervention layer—constitute the

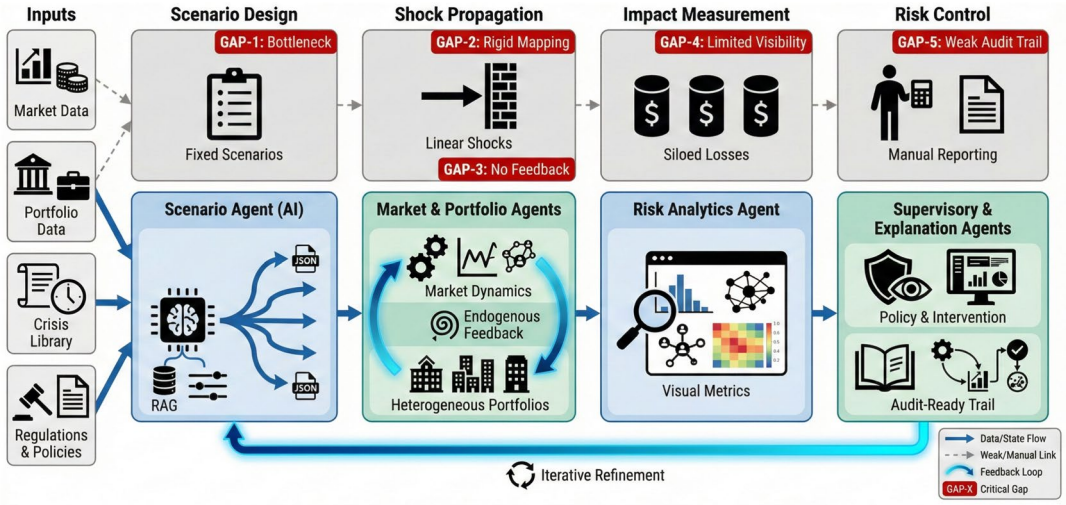


Fig. 1 Gap map: from conventional stress testing to MacroRisk-Agent. Top row: common stress-testing workflows rely on fixed scenarios (GAP-1), simplified linear/exogenous shock mapping (GAP-2) without endogenous feedback (GAP-3), siloed loss measurement with limited system-wide visibility (GAP-4), and manual reporting with weak audit trails (GAP-5). Bottom row: MacroRisk-Agent replaces these steps with a multi-agent pipeline comprising (i) a Scenario Agent (LLM/RAG) that outputs parameterized scenario constraints (e.g., factor-path and liquidity specifications), (ii) interacting Market and Portfolio Agents that capture endogenous feedback and heterogeneous portfolio behaviors, (iii) a Risk Analytics Agent that computes portfolio- and systemic-level metrics, and (iv) Supervisory and Explanation Agents that support policy interventions and produce audit-ready, evidence-linked outputs. Solid arrows denote data/state flow, dashed arrows indicate weak/manual links, and curved arrows represent feedback loops and iterative refinement.

demonstration environment in which this interface is exercised, and are not claimed as independently validated financial-modeling contributions. Concretely:

1. We propose a **RAG-grounded, schema-constrained generation pipeline** with deterministic validation gates (Definitions 1–3) that achieves 0% schema-defined hallucination under the specified validation criteria (Definition 4) (vs. 22.2% without gates) while maintaining 91.4% evidence traceability, providing a principled approach to constrained LLM output generation for high-stakes applications. More precisely, the pipeline reduces the rate of *schema-defined and validation-rule-defined scenario failures* to 0% under the proposed evaluation protocol; it is thus *designed to constrain selected forms of structured-generation failure* through deterministic validation gates, and does not certify that generated narratives are factually correct, that retrieved evidence is complete, or that recommendations are economically sound (see Limitations and the Failure-Case Analysis).
2. We design a **constraint-coupled multi-agent simulation architecture** that explicitly models endogenous feedback through fire-sale, margin, liquidity, and overlapping-holdings loops, producing a 52% higher systemic-loss estimate than the one-pass configuration under the calibrated experimental setting and enabling mechanism-level attribution with 85.4% F1 accuracy.
3. We introduce a **counterfactual re-simulation framework** for intervention validation that achieves 68.6% simulation-defined top-1 decision success rate with 5.6% calibration error, and 83.6% explanation faithfulness, illustrating how an LLM-agent system can be architected to produce auditable, provenance-tracked decision support. We emphasize that "success" here denotes simulation-defined risk reduction under the same engine that generates the dynamics, not demonstrated real-world intervention effectiveness. The faithfulness score measures machine-checkable consistency with internal simulation logs, not economic correctness.

4. We provide an **open-source implementation and reproducible benchmark** on public financial datasets (FRED, Yahoo Finance), with cached LLM outputs and fixed random seeds, enabling independent verification and extension.

Empirically, MacroRisk-Agent achieves statistically significant improvements over the strongest baselines across all three evaluation dimensions: +14.2 percentage points in scenario validity ($p < 0.001$), a 1.52× simulated amplification ratio relative to the one-pass configuration ($p < 0.001$), and +7.2 points in simulation-defined intervention success rate ($p < 0.01$).

Organization

The remainder of the paper is organized as follows. The next section reviews related work on constrained generation with retrieval augmentation, LLM agents for structured decision support, multi-agent simulation systems, and financial stress-testing methodology. We then formalize the problem setup, including the unified system state, scenario representation, and output metrics. The method section presents the MacroRisk-Agent architecture: the scenario generation pipeline with validation gates, market dynamics engine, participant agents with endogenous feedback, risk analytics, and the counterfactual intervention framework. The experimental section specifies datasets, baselines, and evaluation metrics, followed by results across three research questions (RQ1–RQ3). We conclude with limitations and future directions.

Related work

RAG for structured output generation

Retrieval-augmented generation (RAG) augments LLM outputs with retrieved evidence to reduce hallucination and improve factual grounding[5]. Recent surveys document substantial progress in retrieval strategies, chunk optimization, and evaluation frameworks[6], while RAGAS provides automated metrics for retrieval quality and answer faithfulness[12]. On the hallucination detection side, SelfCheckGPT enables zero-resource hallucination detection by measuring consistency across sampled outputs[3]. Complementary work on constrained decoding—such as Outlines[4]—enforces structural validity (e.g., JSON schema compliance) at the token level.

MacroRisk-Agent builds on these foundations but differs in a key respect: our validation gates (Definitions 1–3) enforce *domain-specific* constraints—numerical plausibility, cross-field consistency, and evidence alignment—that go beyond generic schema compliance or statistical hallucination detection. This multi-gate architecture provides layered defense against the distinct failure modes of structured output generation in high-stakes settings.

Constrained generation with retrieval augmentation in scenario design

The stress testing literature spans top-down macro-financial models, institution-level projection frameworks, and hybrid approaches that link macro scenarios to losses[13]. This motivates methodologies that either (i) broaden the scenario set and explicitly characterize tail narratives, or (ii) reduce dependence on a single subjective scenario by adopting alternative stability measures.

A key methodological development is *reverse stress testing*, which seeks to identify scenarios that would break the system (or violate capital/liquidity constraints) rather than evaluating losses under a pre-imposed scenario[14]. formalize scenario design for macroprudential stress tests, illustrating how reverse engineering of "break-the-system" scenarios can systematize tail exploration and reduce the bottleneck of manual scenario selection. Complementary institutional work argues for multi-scenario stress testing to better reflect uncertainty, regime shifts, and non-linearities in tail outcomes[15,16].

These strands collectively motivate the MacroRisk-Agent *Scenario Agent*: scenario design should be (i) more scalable than purely manual workflows, (ii) more tail-aware than single baseline/adverse designs, and (iii) grounded in documented assumptions suitable for supervisory review.

Shock propagation: networks, overlapping portfolios, and price-mediated contagion

A large body of work shows that systemic risk is often driven by *interconnectedness* and *indirect exposures*rather than simple idiosyncratic credit losses. Price-mediated contagion can emerge when institutions with overlapping holdings deleverage simultaneously, producing endogenous price declines, mark-to-market losses, and further deleveraging[9]. develop indicators computable from portfolio holdings and market depth that proxy outcomes of systemic stress tests, highlighting how indirect contagion complements size-based systemic importance measures.

Empirical and quantitative frameworks for overlapping portfolios and fire-sale spillovers have advanced substantially in recent years[10]. quantify systemic risk from indirect interconnections using detailed holdings data and show that ignoring overlapping portfolios can severely underestimate system losses. At the market level[17], provide evidence and modeling of fire-sale spillovers and systemic risk amplification. To connect these channels within stress-testing pipelines[18], propose a tractable "contagion accounting" framework that decomposes system losses into contributions from direct interbank exposures versus indirect fire-sale channels, illustrating that common exposures and fire-sales may dominate loss dynamics in practice.

A complementary literature explores scenario dependence and alternatives to scenario-imposed stress tests[19]. develop an eigenvalue-based approach that enables analysis of interacting contagion channels without requiring a subjective initial stress scenario, addressing a key limitation of scenario-driven system-wide stress tests. While these approaches reduce scenario dependence, their interpretability for policy actions and their integration with portfolio-level risk control workflows remain open challenges–precisely the interface MacroRisk-Agent targets via an agentic scenario generator coupled with auditable shock-to-loss pipelines.

Liquidity stress testing, margin dynamics, and the solvency–liquidity nexus

Liquidity risk has become central to modern stress testing because solvency shocks and liquidity needs are tightly coupled through funding constraints, collateral requirements, and market liquidity[20]. propose a structural framework for *joint* stress testing of solvency and liquidity and introduce "Liquidity at Risk" as a coherent measure of liquidity resources required under stress, showing how funding costs and liquidity needs can amplify equity losses.

At the system-wide level, the IMF has developed tools to operationalize systemwide liquidity stress testing with an emphasis on run dynamics, liquid-asset buffers, and spillovers[21]. In market infrastructures, liquidity demand may be amplified by central clearing and margining practices under stress[22]; analyze central clearing and systemic liquidity risk, highlighting how margin and collateral flows can generate system-wide liquidity strains. These insights align with policy and institutional reports emphasizing liquidity preparedness–particularly for NBFIs facing margin and collateral calls–and the need to monitor procyclical liquidity demands[23,24].

Recent central bank research further documents how liquidity preparedness interacts with market-based finance and NBFI behavior, including collateral usage and margin spirals in stressed markets[25,26]. This literature motivates MacroRisk-Agent's explicit representation of *margin–funding–liquidation* loops and its design choice to treat liquidity constraints as first-class state variables (rather than as an ex post add-on) in systemic and portfolio-level stress testing.

Cross-sector and system-wide stress tests: banks, funds, and endogenous feedback

Cross-sector models integrate banks with asset managers and funds to capture amplification through common asset holdings, market depth, and redemption-driven fire sales. ECB research shows that adding the fund sector to

bank-focused stress tests can materially increase losses via fire sales and capital depletion[27]. More recent system-wide modeling work further quantifies shock amplification in interconnected bank–fund systems using microfinancial and holdings data, reinforcing the importance of joint modeling of fire sales and cross-sector feedback[28].

In addition to measurement, a growing literature addresses *control*: how policy tools, capital buffers, or targeted interventions can reduce contagion losses[29]. provide a model-based calibration approach for additional capital to minimize contagion losses in an interbank system augmented with fire sale mechanisms, illustrating how stress test outputs can be translated into actionable controls. These results motivate MacroRisk-Agent's "Risk Control" layer, which is designed to translate multi-agent simulated dynamics into intervention recommendations and policy-aligned constraints, while keeping a clear provenance for supervisory review.

LLM agents for constrained structured generation and decision support

Parallel to advances in systemic risk modeling, LLMs have rapidly progressed as foundation models for text-heavy financial tasks, enabling domain adaptation and improved performance on financial language understanding. BloombergGPT represents a large language model trained for finance-oriented tasks and benchmarks[30], while open-source efforts such as FinGPT aim to provide adaptable financial LLMs suitable for research and controlled deployment[31]. Regulatory and supervisory communities have begun to assess the opportunities and risks of LLM deployment in finance, emphasizing governance controls, documentation, and human oversight[11].

Methodologically, a growing AI literature proposes tool-using and agentic LLM systems that combine language reasoning with structured actions, multi-agent collaboration, and tool invocation. ReAct formalizes the synergy between reasoning traces and action execution for tool-augmented tasks[1], while AutoGen proposes a general multi-agent conversation framework to build LLM applications with configurable agent roles and interaction patterns[2]. These developments offer a blueprint for translating complex stress testing workflows into modular, auditable agent pipelines: a scenario agent can generate structured scenarios; market and portfolio agents can simulate actions and feedback; an analytics agent can compute risk metrics; and supervisory/explanation agents can produce audit-ready narratives and evidence trails.

However, the finance setting imposes stricter requirements than generic agent applications: hallucination control, constrained outputs, provenance tracking, and policy compliance must be enforced end-to-end. Recent work on LLM-based financial agents, such as FinAgent[32], demonstrates the potential of multimodal LLM agents for financial tasks but does not address the validation-gate architecture or provenance-tracking requirements central to MacroRisk-Agent. Similarly, benchmarks for real-world planning with language agents[33] highlight the gap between LLM capabilities and the constraint-satisfaction demands of structured decision support. These constraints reinforce the design principle of MacroRisk-Agent: LLMs should operate within a *risk-controlled* architecture where (i) scenario generation is grounded via retrieval and validation gates, (ii) simulation and analytics are executed by deterministic engines, and (iii) every artifact is logged and reproducible for audit.

Multi-agent simulation systems

Agent-based modeling (ABM) has a long history in computational economics and financial market simulation. ABIDES provides a high-fidelity multi-agent market simulation environment that models realistic order-book dynamics, latency, and heterogeneous trading strategies[7]. Mesa offers a general-purpose Python framework for agent-based modeling with built-in visualization and data collection[8]. Recent surveys document the growing adoption of ABM at central banks for policy analysis and systemic risk assessment[34,35].

MacroRisk-Agent differs from these systems in a fundamental architectural respect: it *couples LLM-based semantic reasoning with ABM-style numerical simulation*. While ABIDES and Mesa provide flexible simulation frameworks, they require manual specification of scenarios and agent strategies. MacroRisk-Agent automates scenario construction through RAG-grounded generation (Algorithm 2), connects natural-language intent to parameterized simulation inputs via schema-constrained generation, and maintains end-to-end provenance from

retrieved evidence through simulation outcomes to intervention recommendations. This integration of language-model capabilities with multi-agent simulation addresses the semantic-to-numeric coupling challenge (GAP-2) that is absent from purely numerical ABM frameworks.

Positioning against the five gaps

To make the relationship to prior work explicit, Table 1 maps representative streams of the literature reviewed above onto the five technical gaps (GAP-1–GAP-5) and states, for each stream, the specific shortcoming that motivates our design. The recurring pattern is that existing approaches address one or two gaps strongly but leave the others open: systemic-risk models capture feedback (GAP-3/4) but presuppose hand-specified scenarios and provide no provenance layer (GAP-1/5); reverse and multi-scenario stress testing broaden scenario design (GAP-1) but remain manual and are not coupled to behavioral simulation (GAP-2/3); and generic LLM-agent frameworks support flexible orchestration but lack the domain validation gates and end-to-end auditability (GAP-1/5) required in regulated settings. MacroRisk-Agent does not claim to outperform any single specialized model on its own gap; rather, it is the first design we are aware of to address all five gaps within one auditable pipeline, and we are explicit (in the Experimental Results and Limitations) about the asymmetry this creates when comparing against single-channel baselines.

Problem formulation

This section formalizes the stress-testing problem addressed by MacroRisk-Agent. We define the financial entities, market states, scenario representation, agent actions, and the portfolio-level and systemic-level outputs that the system produces. The goal is to support *systemic* and *portfolio-level* stress testing within a unified environment, while enabling auditable, policy-aligned interventions (as illustrated in Fig. 1). For ease of reference, Table 2 (presented up front, at the start of this section) collects all symbols used throughout the paper; readers may consult it while reading the formalization below.

Table 1 Positioning of representative prior work against the five technical gaps. • = addressed; ◦ = partially addressed; – = not addressed. The final column states the shortcoming that MacroRisk-Agent targets.

Prior-work stream (examples)	G1	G2	G3	G4	G5	Key shortcoming addressed here
Reverse/multi-scenario stress design[14,15]	•	–	–			Scenario design is broadened but remains largely manual and is decoupled from behavioral simulation and provenance.
Overlapping-portfolio/fire-sale contagion[10,17,18]	–		•	•	–	Captures a single channel under exogenous scenarios; no constrained generation or audit trail.
Solvency–liquidity/system-wide liquidity[20,21]	–		•	•	–	Strong liquidity dynamics but scenario- and provenance-agnostic; not actionable end-to-end.
Scenario-free contagion analysis[19]		–	•	•	–	Reduces scenario dependence but offers limited policy interpretability and no auditable narrative.
RAG/constrained decoding[3–5]			–	–		Generic schema/faithfulness control without domain plausibility/consistency gates or simulation coupling.
LLM-agent frameworks[1,2,32]				–	–	Flexible orchestration but no deterministic validation gates or end-to-end provenance for regulated use.
ABM market/systemic simulation[7,8,34]	–	–	•	•	–	High-fidelity simulation but requires manual scenario/strategy specification; no semantic-to-numeric interface.
MacroRisk-Agent (this work)	•	•	•	•	•	Single auditable pipeline spanning all five gaps; evaluated as a proof-of-concept (see Limitations).

Table 2 Notation and definitions used in the problem setup.

Notation	Definition
$\mathcal{T} = \{0, \dots, T\}$	Discrete time index set and horizon length T.
$\mathcal{I}$	Set of institutions/portfolios (banks, funds, dealers, desks).
$\mathcal{A}$	Asset universe (cash instruments and derivatives as modeled).
$\mathcal{F}$	Risk-factor set (rates, spreads, FX, equities, volatility, etc.).
$s \in \mathcal{S}$	Stress scenario indexed by s.
N_s	Narrative component of scenario s (human-readable story).
Ω_s	Parameterized constraint set of scenario s (admissible factor/liquidity dynamics).
$\mathrm{x}_t \in \mathbb{R}^{\lvert\mathcal{F}\rvert}$	Risk-factor vector at time t.
$\mathrm{p}_t \in \mathbb{R}^{\lvert\mathcal{A}\rvert}$	Asset price vector at time t (mapped from x_t).
$g(\cdot)$	Pricing/risk-factor-to-asset mapping (may include valuation models).
ℓ_t	Liquidity-state vector (e.g., spreads, depth, impact parameters).
$h(\cdot)$	Liquidity transition/update function under stress and order flow.
$\mathrm{w}_{i,t} \in \mathbb{R}^{\lvert\mathcal{A}\rvert}$	Holdings/positions of institution i at time t.
$\mathrm{u}_{i,t} \in \mathbb{R}^{\lvert\mathcal{A}\rvert}$	Action (trade/hedge/deleveraging) of institution i at time t.
$\mathrm{q}_t = \sum_{i \in \mathcal{I}} \mathrm{u}_{i,t}$	Aggregate net order flow across institutions at time t.
$V_{i,t}$	Market value of institution i's portfolio at time t.
$L_{i,t}$	Cumulative loss of institution i up to time t (scenario-path dependent).
$L_{i,T}$	Terminal loss of institution i at horizon T.
$L_T^{\mathrm{sys}} = \sum_i L_{i,T}$	System aggregate terminal loss at horizon T.
$\mathrm{VaR}_\alpha(\cdot)$, $\mathrm{ES}_\alpha(\cdot)$	Value-at-Risk and Expected Shortfall at confidence level α.
$z_{i,t}$	Auxiliary institution variables (capital, leverage, collateral, funding gaps, limit usage).
$\mathcal{C}_i$	Feasibility/constraint set for institution i (risk limits, leverage, margin, funding).
$\mathcal{U}_i(\mathrm{s}_t)$	Set of feasible actions for i given current system state s_t.
s_t	Unified system state at time t: $(\mathrm{x}_t, \ell_t, \{\mathrm{w}_{i,t}, z_{i,t}\}_{i \in \mathcal{I}})$.
$\mathcal{T}_s(\cdot)$	Scenario-constrained transition operator mapping s_t to s_{t+1}.
θ	Collection of model parameters (factor dynamics, liquidity, pricing, behavior).
$\mathcal{A}^{\mathrm{ctl}}$	Set of risk control actions (capital buffers, hedging directions, limit changes).
$\mathcal{F}^{\mathrm{ctl}}$	Feasibility constraints for control actions (policy, cost, operational constraints).
$\Phi(\cdot)$	Stress outcome objective aggregator (systemic + portfolio tail risk penalties).
$\mathrm{Cost}(a)$	Implementation/opportunity cost of intervention a.
λ	Trade-off weight between risk reduction and intervention cost.

Entities, markets, and time

We consider a discrete-time horizon indexed by $t \in \mathcal{T} = \{0, 1, \dots, T\}$. The financial system contains a set of institutions/portfolios $\mathcal{I}$ (e.g., banks, funds, dealers) that hold and trade assets from a universe $\mathcal{A}$. Market conditions are summarized by a set of risk factors $\mathcal{F}$, such as interest-rate curve nodes, credit-spread factors, FX rates, equity indices, and volatility factors.

At each time t, the system state consists of (i) market and liquidity states, and (ii) institution/portfolio states. MacroRisk-Agent simulates the evolution of these states under stress scenarios and endogenous behavioral responses.

Market state and liquidity state

Let $\mathrm{x}_t \in \mathbb{R}^{\lvert\mathcal{F}\rvert}$ denote the vector of risk-factor levels at time t. A mapping $g(\cdot)$ converts risk factors into asset prices:

$$\mathrm{p}_t = g(\mathrm{x}_t), \qquad \mathrm{p}_t \in \mathbb{R}^{\lvert\mathcal{A}\rvert}, \tag{1}$$

where $\mathrm{p}_t[a]$ is the (mid) price of asset $a \in \mathcal{A}$.

To capture liquidity effects, we introduce a liquidity-state vector ℓ_t that encodes market depth, bid–ask spreads, and price-impact parameters. For example, $\ell_t[a]$ may represent the depth of asset a, or a tuple $(\text{spread}_t[a], \text{depth}_t[a], \text{impact}_t[a])$. Liquidity conditions evolve with stress and trading flows:

$$\ell_{t+1} = h(\ell_t, \mathrm{q}_t, \mathrm{x}_t; \theta_\ell), \tag{2}$$

where q_t aggregates net order flow across institutions and θ_ℓ denotes liquidity-model parameters.

Institution/portfolio state and constraints

Each institution $i \in \mathcal{I}$ holds a portfolio $\mathrm{w}_{i,t} \in \mathbb{R}^{|\mathcal{A}|}$, where $w_{i,t}[a]$ denotes the position size (shares/units or notional) in asset a. Let $V_{i,t}$ denote portfolio market value and $\Pi_{i,t}$ denote profit-and-loss (P&L) increments.

Institutions operate under funding, margin, and risk-limit constraints. We represent a general constraint set

$$\mathcal{C}_i \triangleq \left\{(\mathrm{w}_{i,t}, \mathrm{u}_{i,t}) \mid \text{Leverage}_i(\cdot) \le \bar{\lambda}_i,\ \text{Margin}_i(\cdot) \le M_{i,t},\ \text{Risk}_i(\cdot) \le \bar{R}_i,\ \text{LiquidityNeed}_i(\cdot) \le B_{i,t}\right\}, \tag{3}$$

where $\mathrm{u}_{i,t}$ is the institution's action (trade/hedge/deleveraging decision) at time t, and $(\bar{\lambda}_i, \bar{R}_i)$ are leverage and risk limits. $M_{i,t}$ denotes available collateral/margin capacity and $B_{i,t}$ denotes liquid-asset buffers or funding capacity.

Trading affects both institution states and market/liquidity states. We write the institution state update generically as

$$\mathrm{w}_{i,t+1} = \mathrm{w}_{i,t} + \mathrm{u}_{i,t}, \qquad \mathrm{u}_{i,t} \in \mathcal{U}_i(\mathrm{s}_t), \tag{4}$$

where s_t is the full system state (defined next) and $\mathcal{U}_i(\mathrm{s}_t)$ denotes feasible actions given current constraints, prices, and liquidity.

Stress scenario representation

A stress scenario $s \in \mathcal{S}$ is represented as a pair:

$$s \triangleq (N_s, \Omega_s), \tag{5}$$

where N_s is a *narrative* description (human-readable) and Ω_s is a *parameterized constraint set* that specifies admissible stress dynamics. Concretely, Ω_s may include:

- **Factor-path constraints:** bounds and shapes for x_t over $t = 0..T$ (e.g., terminal shocks, path volatility, drawdown limits, regime-switch flags, correlation lift parameters);
- **Liquidity degradation rules:** how ℓ_t deteriorates under stress and order flow (e.g., spread multipliers, depth shrinkage, impact amplification);
- **Policy reaction constraints (optional):** admissible interventions such as liquidity injections or rate paths, if the scenario includes policy responses.

MacroRisk-Agent's Scenario Architect Agent produces (N_s, Ω_s), typically outputting Ω_s in a structured schema (e.g., JSON) that is auditable and versioned.

Unified system state and transition dynamics

Define the unified system state at time t as

$$\mathrm{s}_t \triangleq \left(\mathrm{x}_t, \ell_t, \{\mathrm{w}_{i,t}, z_{i,t}\}_{i \in \mathcal{I}}\right), \tag{6}$$

where $z_{i,t}$ denotes auxiliary institution variables (capital, leverage, collateral, funding gaps, limit utilization, etc.).

Given a scenario s and agent actions $\{u_{i,t}\}$, the state evolves according to a scenario-constrained transition operator:

$$s_{t+1} = \mathcal{T}_s\left(s_t, \{u_{i,t}\}_{i\in\mathcal{I}}; \theta\right) \quad \text{subject to} \quad (x_{0:T}, \ell_{0:T}) \in \Omega_s, \tag{7}$$

where θ collects model parameters for factor dynamics, pricing, and liquidity.

This representation allows MacroRisk-Agent to (i) enforce scenario constraints (severe yet plausible stress evolution) and (ii) capture endogenous feedback through the dependence of market/liquidity updates on aggregated actions $q_t = \sum_i u_{i,t}$.

Outputs: portfolio-level and systemic-level stress metrics

For each institution/portfolio i, we compute portfolio-level risk outcomes under scenario s, including:

- **Loss trajectory:** $L_{i,t}$ and terminal loss $L_{i,T}$;
- **Tail risk:** $\mathrm{VaR}_\alpha(L_{i,T})$ and $\mathrm{ES}_\alpha(L_{i,T})$ over scenario Monte Carlo paths;
- **Drawdowns and attribution:** maximum drawdown and factor/asset contribution measures.

At the system level, we compute systemic risk indicators such as:

- **System loss/tail loss:** aggregate losses $L_T^{\mathrm{sys}} = \sum_i L_{i,T}$ and tail measures;
- **Default/constraint-violation statistics:** number of defaults or binding-constraint events, clustering measures;
- **Contagion and liquidity stress:** spillover depth proxies, liquidity shortfalls, and market-functioning degradation measures derived from ℓ_t and forced sales.

Risk control actions and objective

Beyond measurement, MacroRisk-Agent supports a *risk control loop*: based on stress outcomes, a policy layer proposes interventions (capital buffers, de-risking priorities, hedging directions, limit adjustments) and evaluates their counterfactual effects via re-simulation. Let $a \in \mathcal{A}^{\mathrm{ctl}}$ denote a control action (or action bundle), with feasibility constraints $\mathcal{F}^{\mathrm{ctl}}$. We define a generic intervention objective:

$$\min_{a\in\mathcal{F}^{\mathrm{ctl}}} \underbrace{\mathbb{E}\left[\Phi\left(L_T^{\mathrm{sys}}, \{L_{i,T}\}_{i\in\mathcal{I}}\right)\right]}_{\text{stress outcomes}} + \underbrace{\lambda \cdot \mathrm{Cost}(a)}_{\text{implementation / opportunity cost}}, \tag{8}$$

where $\Phi(\cdot)$ summarizes systemic and portfolio tail risk (e.g., weighted ES, default penalties, liquidity-shortfall penalties) and $\mathrm{Cost}(a)$ captures capital, liquidity, or business costs of interventions. In practice, MacroRisk-Agent treats the control problem as a decision-support task: it proposes actions, generates auditable rationales, and validates improvements via counterfactual stress runs. All symbols used in this section and the remainder of the paper are collected in Table 2 above.

Having formalized the stress-testing problem, including the unified system state, scenario representation, and portfolio/systemic objectives that MacroRisk-Agent targets, we now present the system architecture and algorithmic components that operationalize this framework. The following section describes how MacroRisk-Agent addresses each of the five gaps (GAP-1 through GAP-5) identified in Fig. 1 through a coordinated multi-agent design.

Proposed MacroRisk-Agent method

MacroRisk-Agent is a unified stress-testing and risk-control system that combines (i) constrained scenario construction grounded in crisis and regulatory knowledge, (ii) multi-agent simulation of market-wide propagation with endogenous feedback, (iii) joint portfolio- and systemic-risk analytics, and (iv) audit-ready reporting and governance support. The design is motivated by the limitations summarized in Fig. 1: conventional workflows often rely on a small fixed scenario set (GAP-1), map narrative shocks to portfolio outcomes through rigid and largely exogenous paths (GAP-2), omit endogenous behavioral feedback (GAP-3), offer limited system-wide visibility (GAP-4), and produce weak end-to-end audit trails that separate measurement from management action (GAP-5). Figure 2 instantiates our response as a layered architecture centered on a shared financial environment that is interacted with by specialized agents, while Figs. 3–5 detail three core mechanisms: (i) narrative-to-constraints translation with validation gates, (ii) endogenous amplification loops (fire sales, margin spirals, liquidity spirals, overlapping holdings), and (iii) a counterfactually validated intervention loop for risk control.

Design rationale and end-to-end workflow

A central design choice is to separate *semantic* tasks from *numerical* tasks while keeping them tightly coupled through a structured scenario interface. MacroRisk-Agent uses a Fin-LLM with retrieval-augmented generation (RAG) to synthesize information from a crisis library and a regulatory/policy base, but it does not delegate numerical simulation or risk computation to the language model. Instead, the LLM produces a parameterized constraint object Ω_s (for example, admissible ranges for rate and spread paths, dependence shifts such as correlation lift, and liquidity-degradation multipliers), which is then consumed by quantitative engines to generate reproducible factor paths, prices, and liquidity states. This "LLM-to-constraints" separation is essential for auditability: it allows the system to exploit LLM strengths in narrative structuring and policy-text interpretation, while maintaining deterministic, testable pipelines for valuation, simulation, and metric computation[30,31].

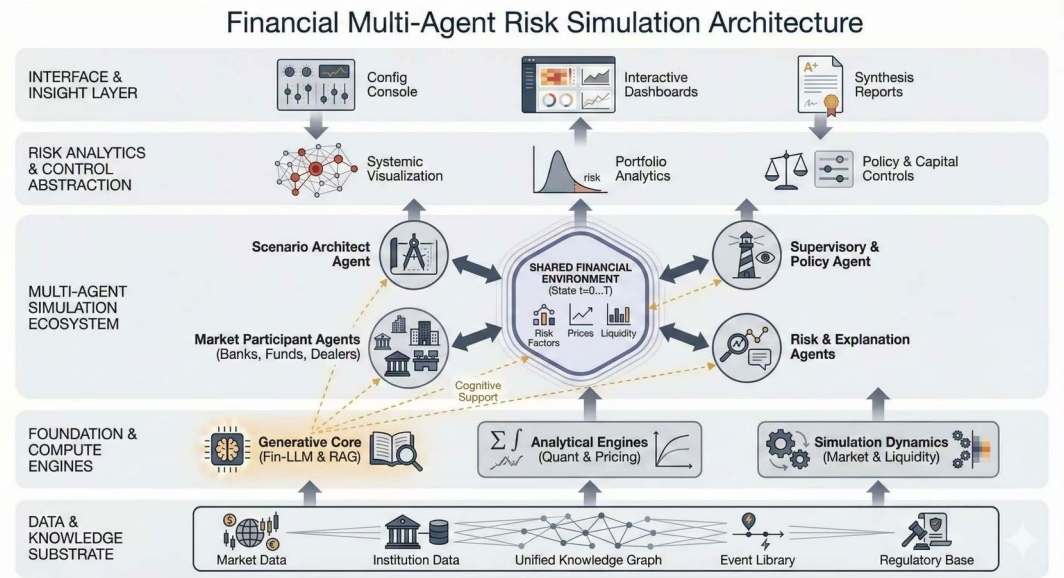


Fig. 2 MacroRisk-Agent architecture ("Financial Multi-Agent Risk Simulation Architecture"). The system integrates a data & knowledge substrate (market and institution data, unified knowledge graph, event library, regulatory base), foundation and compute engines (Fin-LLM with retrieval, analytical engines for pricing and risk, and market/liquidity simulation dynamics), a multi-agent simulation core operating over a shared financial environment (risk factors, prices, liquidity), and an interface layer for configuration, dashboards, policy controls, and audit-ready reports.

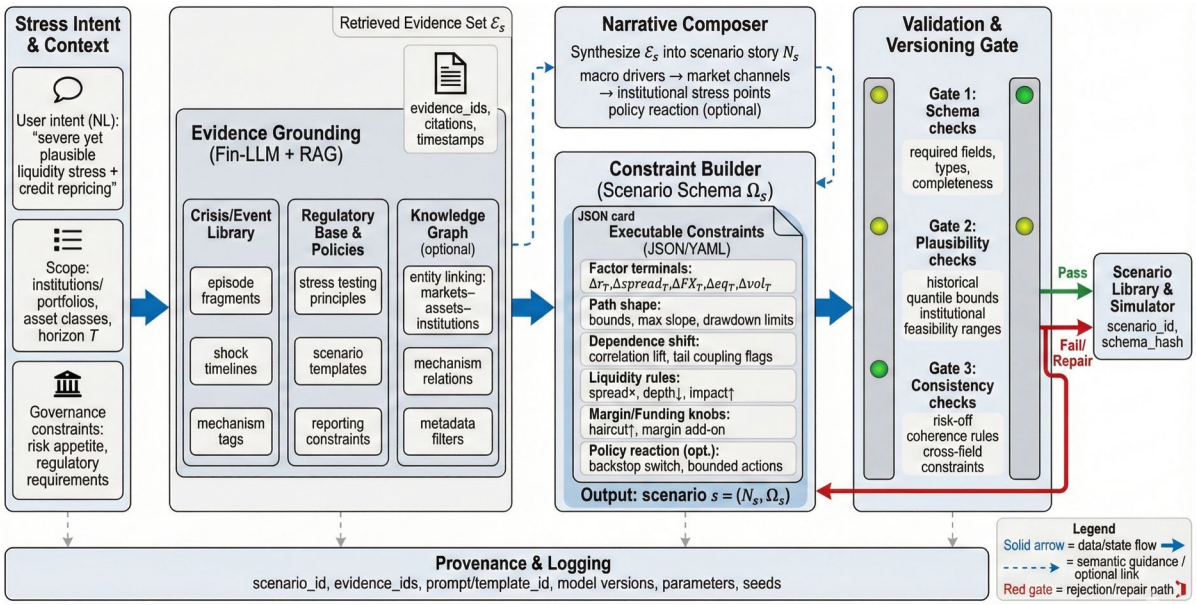


Fig. 3 Narrative-to-constraints translation with validation gates. The Scenario Architect Agent grounds scenario intent using retrieval (RAG) from crisis episodes, regulatory/policy texts, and a knowledge substrate, producing an evidence set $\mathcal{E}_s$. A narrative N_s is synthesized for human interpretability, while an executable constraint schema Ω_s specifies factor-path bounds, dependence shifts, and liquidity degradation rules. Deterministic validation gates enforce schema completeness, numerical plausibility, and cross-field consistency; passing scenarios are versioned and stored with provenance metadata, while failed scenarios are repaired or rejected.

Operationally, MacroRisk-Agent executes each stress-testing session as a closed loop. The user begins at the interface layer (top of Fig. 2) by specifying the scope (institutions/portfolios, asset classes, horizon) and the objective (e.g., "severe-yet-plausible liquidity stress with rapid spread widening and margin spirals") together with governance constraints such as risk appetite targets or supervisory requirements. The Scenario Architect Agent then retrieves relevant historical episodes and policy snippets and generates a narrative N_s together with a structured constraint set Ω_s.

Given (N_s, Ω_s), the simulation core initializes the shared financial environment at $t = 0$ using the market and institution data substrate. Over time steps $t = 0, \ldots, T-1$, the market dynamics engine generates factor and liquidity evolution consistent with Ω_s, while market participant agents representing banks, funds, and dealers respond to binding constraints—risk limits, leverage, margin, and funding needs—by adjusting positions. Their actions aggregate into net order flows that feed back into liquidity and prices, producing endogenous amplification when many agents attempt to de-risk simultaneously.

After completing the horizon (or after reaching absorbing events such as default/binding liquidity failures), the Risk Analytics Agent computes portfolio and systemic metrics, and the Supervisory & Policy Agent proposes intervention bundles (capital buffers, de-risking priorities, hedging directions, and limit adjustments). Recommendations are not treated as one-shot prescriptions: MacroRisk-Agent validates candidate actions through counterfactual re-simulation, generating before/after comparisons under identical scenario constraints and documenting both expected impact and trade-offs.

Finally, the Risk & Explanation Agents produce an evidence-linked narrative report with a provenance record that includes scenario identifiers, retrieved evidence, model versions, and simulation seeds, enabling the kind of audit-ready traceability sought in supervisory-style review[11,36].

Algorithm 1 summarizes the end-to-end loop. Its five core stages concretize the design principles: the ScenarioArchitect translates objectives into an auditable constraint set Ω_s (Line 1); the simulation loop couples market dynamics with heterogeneous agent behaviors, enabling endogenous feedback (Lines 2–9); RiskAnalytics

computes unified portfolio and systemic metrics (Line 10); SupervisoryPolicy and CounterfactualEval together generate and validate actionable interventions (Lines 11–12); and the Explanation agent produces an audit-ready report (Line 13). This closed-loop process directly addresses the challenges of scenario rigidity, under-modeled feedback, and weak auditability outlined in the Introduction.

Require: Data/knowledge substrate $\mathscr{D}$; institutions/portfolios $\mathscr{I}$; objectives/constraints $\mathscr{G}$; horizon T
Ensure: Scenario $s = (N_s, \Omega_s)$; metrics $\mathscr{M}$; interventions $a^\star$; counterfactual impact $\Delta\mathscr{M}$; audit-ready report R
1: $s \leftarrow \textsc{ScenarioArchitect}(\mathscr{G}, \mathscr{D})$ // RAG-grounded, schema-constrained
2: Initialize $\mathbf{s}_0$ from $\mathscr{D}$
3: **for** $t = 0$ **to** $T - 1$ **do**
4: $\quad \mathbf{x}_{t+1}, \ell_{t+1} \leftarrow \textsc{MarketDynamics}(\mathbf{s}_t, \Omega_s)$
5: $\quad$ **for all** $i \in \mathscr{I}$ **do**
6: $\quad\quad \mathbf{u}_{i,t} \leftarrow \textsc{ParticipantAgent}_i(\mathbf{s}_t, \Omega_s)$
7: $\quad$ **end for**
8: $\quad \mathbf{s}_{t+1} \leftarrow \mathscr{T}_s(\mathbf{s}_t, \{\mathbf{u}_{i,t}\}_{i \in \mathscr{I}})$
9: **end for**
10: $\mathscr{M} \leftarrow \textsc{RiskAnalytics}(\{\mathbf{s}_t\}_{t=0}^{T})$
11: $a^\star \leftarrow \textsc{SupervisoryPolicy}(\mathscr{M}, \mathscr{G}, \mathscr{D})$
12: $\Delta\mathscr{M} \leftarrow \textsc{CounterfactualEval}(a^\star)$ // re-simulate under same Ω_s
13: $R \leftarrow \textsc{Explanation}(\mathscr{M}, a^\star, \Delta\mathscr{M}, s, \mathscr{D})$
14: **return** $(s, \mathscr{M}, a^\star, \Delta\mathscr{M}, R)$

Algorithm 1 MacroRisk-Agent end-to-end stress testing and control loop.

Constrained scenario generation: from intent to executable constraints

The Scenario Architect Agent addresses GAP-1 (scenario bottleneck) by transforming high-level stress intent into an auditable, parameterized scenario representation that supports both diversity and control. Rather than producing only a narrative, the agent emits a pair $s = (N_s, \Omega_s)$, where N_s is the human-readable stress story and Ω_s is a machine-executable constraint specification that binds factor paths, dependence shifts, and liquidity degradation rules. This representation aligns with the broader move toward systematic scenario design, including reverse stress testing and multi-scenario tail exploration[14,15]. Algorithm 2 details the generation pipeline.

Require: Governance constraints $\mathcal{G}$, data/knowledge substrate $\mathcal{D}$, max retries K, temperatures τ_N=0.7, τ_Ω=0, similarity threshold δ
Ensure: Validated scenario $s = (N_s, \Omega_s)$ or `null`

$\mathcal{E}_s \leftarrow \textsc{Retrieve}(\mathcal{G}, \mathcal{D}, k{=}20, \text{MMR}, \delta)$ // RAG: top-k with MMR re-ranking
$N_s \leftarrow \text{LLM}(\text{Prompt}_N, \mathcal{E}_s, \mathcal{G}, \tau_N)$ // Generate narrative (Prompt S1)
$\Omega_s^{\text{raw}} \leftarrow \text{LLM}(\text{Prompt}_\Omega, N_s, \mathcal{E}_s, \mathcal{G}, \tau_\Omega)$ // Generate schema (Prompt S2, JSON mode)
for $r = 1$ **to** K **do**
  $v_1 \leftarrow \textsc{GateSchema}(\Omega_s^{\text{raw}})$ // Def. 1
  if not v_1.pass **then**
    $\Omega_s^{\text{raw}} \leftarrow \text{LLM}(\text{Prompt}_{\text{repair}}, \Omega_s^{\text{raw}}, v_1.\text{errors})$; **continue**
  end if
  $v_2 \leftarrow \textsc{GatePlausibility}(\Omega_s^{\text{raw}}, \mathcal{D})$ // Def. 2
  if not v_2.pass **then**
    $\Omega_s^{\text{raw}} \leftarrow \text{LLM}(\text{Prompt}_{\text{repair}}, \Omega_s^{\text{raw}}, v_2.\text{errors})$; **continue**
  end if
  $v_3 \leftarrow \textsc{GateConsistency}(\Omega_s^{\text{raw}})$ // Def. 3
  if not v_3.pass **then**
    $\Omega_s^{\text{raw}} \leftarrow \text{LLM}(\text{Prompt}_{\text{repair}}, \Omega_s^{\text{raw}}, v_3.\text{errors})$; **continue**
  end if
  $\Omega_s \leftarrow \Omega_s^{\text{raw}}$; **break** // All gates passed
end for
if $r > K$ **then**
  return `null` // Reject after max retries
end if
$\text{id}_s \leftarrow \textsc{Hash}(\Omega_s \| \mathcal{E}_s.\text{ids} \| \text{timestamp})$
$\textsc{Store}(s, \text{id}_s, \mathcal{E}_s.\text{ids}, \text{model_ver}, \text{seeds})$ // Provenance logging
return (N_s, Ω_s)

Algorithm 2 Scenario Architect: RAG-grounded constrained generation with validation gates.

Figure 3 details the narrative-to-constraints pipeline. The process begins with evidence grounding: given the user's objective and governance constraints, the agent retrieves an evidence set $\mathcal{E}_s$ from the crisis/event library, regulatory base, and (optionally) a knowledge graph linking markets, institutions, and mechanisms. The retrieved evidence includes episode fragments (shock sequencing and timing), magnitude cues (historical ranges and regime behavior), and governance constraints (e.g., "severe yet plausible", reporting templates). The LLM then synthesizes $\mathcal{E}_s$ into a coherent narrative N_s, but importantly, it also fills a structured schema for Ω_s. The schema includes terminal and path constraints for key risk factors (rates, spreads, FX, equities, volatility), dependence adjustments (e.g., correlation lift during stress), and liquidity multipliers specifying spread widening, depth deterioration, and impact amplification. By requiring Ω_s to be explicit and bounded, the downstream engines never need to interpret free-form text to produce numbers, which is crucial for reproducibility and governance[11].

To prevent uncontrolled scenario generation, MacroRisk-Agent applies a multi-stage validation gate before simulation begins. We formalize the three gate conditions:

Definition 1 *(Schema Completeness Gate)* A scenario constraint set Ω_s passes the schema completeness gate if and only if Ω_s contains all fields in the required field set $\mathcal{F}_{\text{req}}$ (terminal factor shocks, path volatility bounds, correlation specification, and liquidity degradation parameters), each field conforms to its specified type, and all numerical ranges are non-empty.

Definition 2 *(Plausibility Gate)* A scenario constraint set Ω_s passes the plausibility gate if and only if, for every risk factor $f \in \mathcal{F}$, the terminal shock magnitude satisfies $|\Delta x_T(f)| \le Q_{99}(|\Delta x_T(f)|; \mathcal{D}_{\text{hist}})$, where Q_{99} denotes the 99th

percentile of historical factor changes computed from the calibration dataset $\mathcal{D}_{\text{hist}}$. This enforces the "severe yet plausible" requirement.

Definition 3 *(Consistency Gate)* A scenario constraint set Ω_s passes the consistency gate if and only if the following risk-off coherence rules hold: (C1) equity shock $< -15\%$ implies volatility shock > 0; (C2) credit spread shock > 100 bps implies liquidity depth multiplier < 0.9; (C3) the scenario correlation matrix Σ_s is positive definite; and (C4) path-speed constraints are consistent with terminal constraints (i.e., the terminal value is reachable within the specified speed limits over horizon T).

Definition 4 *(Schema Hallucination and Hallucination Rate)* A generated scenario Ω_s exhibits *schema hallucination* if any of the following conditions hold: (i) a field value exceeds the historical 99.9th percentile without supporting evidence from the retrieved set $\mathcal{E}_s$; (ii) cross-field relationships violate rules C1–C4 in Definition 3; or (iii) the maximum evidence alignment score between Ω_s fields and retrieved fragments falls below a threshold δ_{ev}. The *hallucination rate* is defined as

$$\text{HallucinationRate} = \frac{|\{s : \text{Hallucinated}(s)\}|}{|\{s : \text{AllGenerated}\}|}. \tag{9}$$

Scope and self-referential caveat. Definition 4 captures a deliberately narrow, *schema-level* notion of hallucination: a scenario is flagged only if it violates schema completeness, exceeds historical bounds without evidence, breaches consistency rules C1–C4, or falls below an evidence-alignment threshold. It does *not* certify the absence of broader financial, causal, or explanatory error—a scenario can be schema-valid yet economically misleading. Moreover, because accepted scenarios are exactly those that pass the same gates used to define hallucination, the gates cannot fail on accepted scenarios *by construction*; the headline "0% schema-defined hallucination under the specified validation criteria" Fig. therefore measures *filter consistency*, not independently verified correctness. An independent assessment (expert review of accepted scenarios, or comparison against ground-truth crisis parameters) is required to substantiate a stronger claim; we discuss this explicitly in the Limitations and report an illustrative manual spot-check in the failure-case analysis below.

Failed scenarios are repaired via an LLM repair prompt (see Supplementary Material, Prompt S3) or rejected after K retries and routed back to schema construction; accepted scenarios are stored in a scenario library with a version identifier, schema hash, evidence identifiers, and generation configuration (template id, model version, randomization settings). This yields an auditable scenario supply that can be systematically expanded to fill coverage gaps detected by analytics.

Market dynamics engine: constrained factor paths and endogenous liquidity states

The market dynamics engine addresses GAP-2 (rigid shock mapping) by translating Ω_s into scenario-consistent trajectories of risk factors, asset prices, and liquidity state variables. Let x_t denote the risk-factor vector and ℓ_t denote liquidity state (spreads, depth, impact parameters). We model factor evolution as a scenario-conditioned process

$$\mathrm{x}_{t+1} = \mu_s(t, \mathrm{x}_t) + \Sigma_s(t, \mathrm{x}_t)\,\epsilon_{t+1}, \qquad \epsilon_{t+1} \sim \mathcal{N}(0, I), \tag{10}$$

where μ_s and Σ_s embed scenario-dependent regimes (e.g., stressed volatility, correlation lift) and are calibrated from baseline periods and crisis-like regimes when available. Constraints in Ω_s–terminal shocks, path bounds, speed limits, and dependence adjustments–are enforced through conditional simulation (accept/reject with repair), penalty-based projection onto admissible sets, or reweighting of Monte Carlo paths. The implementation is modular; what matters is that the resulting paths satisfy $(\mathrm{x}_{0:T}, \ell_{0:T}) \in \Omega_s$ and are reproducible under fixed seeds.

Liquidity dynamics are modeled explicitly because liquidity is the key amplification channel linking market-wide behavior to portfolio outcomes. We update liquidity state as

$$\ell_{t+1} = h(\ell_t, \mathrm{q}_t, \mathrm{x}_t; \theta_\ell, \Omega_s), \tag{11}$$

where q_t is aggregate net order flow across agents and $h(\cdot)$ encodes both scenario-imposed deterioration (via Ω_s multipliers) and endogenous deterioration due to trading pressure. Concretely, the liquidity update for each asset a uses a linear price-impact model with nonlinear liquidity degradation:

$$\mathrm{impact}_t(a) = \frac{\kappa \cdot |q_t(a)|}{\mathrm{depth}_t(a)}, \tag{12}$$

$$\mathrm{spread}_{t+1}(a) = \mathrm{spread}_t(a) \cdot \exp\Big(\eta_s \cdot \frac{|q_t(a)|}{\mathrm{depth}_t(a)}\Big) \cdot \omega_s^{\mathrm{spread}}, \tag{13}$$

$$\mathrm{depth}_{t+1}(a) = \mathrm{depth}_t(a) \cdot \exp\Big(-\eta_d \cdot \frac{|q_t(a)|}{V_t^{\mathrm{mkt}}}\Big) \cdot \omega_s^{\mathrm{depth}}, \tag{14}$$

where κ is a price-impact coefficient, η_s and η_d are calibratable elasticities governing how order flow degrades spreads and depth, V_t^{mkt} is total market volume, and $\omega_s^{\mathrm{spread}}, \omega_s^{\mathrm{depth}} \in \Omega_s$ are scenario-specific stress multipliers. The exponential form ensures positivity and captures the empirical observation that liquidity deterioration accelerates nonlinearly under stress[37]. Parameters (κ, η_s, η_d) are calibrated from bid-ask spread and order-book depth data during the 2008 GFC and 2020 COVID-19 periods.

This structure is consistent with joint solvency–liquidity stress testing and system-wide liquidity stress perspectives[20,21,23]. By exposing ℓ_t as a first-class state variable that responds to both stress severity and collective selling, MacroRisk-Agent can represent liquidity spirals and price-impact amplification rather than treating liquidity as static haircuts.

Multi-agent constraint-coupled simulation with emergent feedback

MacroRisk-Agent models market participant behavior using a population of institution/portfolio agents representing banks, funds, and dealers (Fig. 2). Each agent $i \in \mathcal{I}$ holds a portfolio $\mathrm{w}_{i,t}$ and chooses actions $\mathrm{u}_{i,t}$ in response to the current state s_t and scenario constraints Ω_s:

$$\mathrm{u}_{i,t} = \pi_i(\mathrm{s}_t, \Omega_s; \varphi_i) \quad \text{subject to} \quad (\mathrm{w}_{i,t}, \mathrm{u}_{i,t}) \in \mathcal{C}_i, \tag{15}$$

where $\mathcal{C}_i$ encodes risk limits, leverage constraints, margin/collateral requirements, and funding/liquidity buffers. The objective is to capture empirically grounded and policy-relevant mechanisms through which constraints generate procyclical de-risking and market-wide spillovers.

Agent behavior is implemented as constrained responses to trigger events, formalized in Algorithm 3. A priority-ordered trigger cascade determines the agent's action at each time step: leverage breaches (T1) take highest priority, followed by margin calls (T2), risk-limit violations (T3), and liquidity-buffer depletion (T4). Each trigger invokes a specialized subroutine that computes the minimum adjustment required to restore constraint compliance, subject to the current liquidity state ℓ_t. When no trigger fires, the agent takes no action ($\mathrm{u}_{i,t} = 0$). This rule-based design aligns with the literature on price-mediated contagion, overlapping-portfolio spillovers, and fire-sale amplification[10,17,18] and with policy concerns around margin and collateral calls in stressed markets[23].

Require: System state $\mathbf{s}_t$, scenario constraints Ω_s, agent parameters $(\bar{\lambda}_i, \bar{R}_i, \bar{B}_i)$
Ensure: Action $\mathbf{u}_{i,t} \in \mathscr{U}_i(\mathbf{s}_t)$
Compute leverage: $\text{lev}_{i,t} \leftarrow V_{i,t}/(M_{i,t} + B_{i,t})$
Compute margin gap: $\text{mg}_{i,t} \leftarrow \max\big(0,\ \text{MarginReq}(\mathbf{w}_{i,t}, \text{vol}_t) - M_{i,t}\big)$
Compute VaR: $\text{VaR}_{i,t} \leftarrow \text{VaR}(\mathbf{w}_{i,t}, \mathbf{x}_t, \Sigma_t, \alpha{=}0.99)$
if $\text{lev}_{i,t} > \bar{\lambda}_i$ **then**
  return $\textsc{Deleverage}(\mathbf{w}_{i,t}, \text{lev}_{i,t}, \bar{\lambda}_i, \ell_t)$ // T1: Leverage
else if $\text{mg}_{i,t} > 0$ **then**
  return $\textsc{MeetMargin}(\mathbf{w}_{i,t}, \text{mg}_{i,t}, \ell_t)$ // T2: Margin
else if $\text{VaR}_{i,t} > \bar{R}_i$ **then**
  return $\textsc{ReduceRisk}(\mathbf{w}_{i,t}, \text{VaR}_{i,t}, \bar{R}_i, \ell_t)$ // T3: Risk
else if $B_{i,t} < \bar{B}_i$ **then**
  return $\textsc{RaiseLiquidity}(\mathbf{w}_{i,t}, \bar{B}_i - B_{i,t}, \ell_t)$ // T4: Liquidity
end if
return 0 // No action needed

Algorithm 3 Participant agent policy: $scParticipantAgent_i(\mathrm{s}_t, \Omega_s)$

The DELEVERAGE subroutine sells positions proportional to $w_{i,t}[a] \cdot \ell_t[a]^{-1}$, prioritizing assets with high VaR contribution and low liquidity. MEETMARGIN liquidates the minimal-cost portfolio slice to cover the margin gap, considering transaction costs from ℓ_t. REDUCERISK reduces exposure to factors with the highest marginal VaR contribution. RAISELIQUIDITY sells liquid assets first (those with the smallest bid-ask spreads) and resorts to illiquid sales only if buffers are exhausted.

Figure 4 summarizes the endogenous feedback loops that MacroRisk-Agent explicitly represents. Fire-sale dynamics arise when losses and binding constraints induce forced selling, which depresses prices and increases mark-to-market losses, triggering further de-risking. Margin spirals emerge when volatility and haircuts increase margin requirements, producing collateral shortfalls that force liquidation, which further increases volatility and stress. Liquidity spirals operate through market depth and spreads: deteriorating liquidity raises execution costs and slippage, which accelerates losses and incentivizes faster liquidation, further eroding depth.

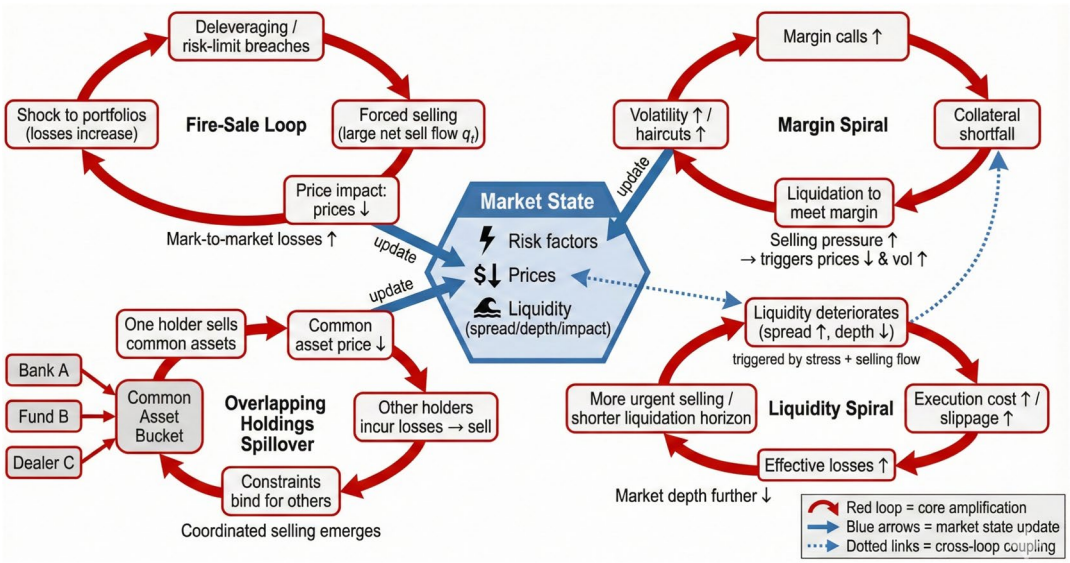


Fig. 4 Endogenous feedback loops and amplification channels modeled in MacroRisk-Agent. The system captures price-mediated contagion mechanisms including fire-sale loops (forced selling → price impact → losses → further selling), margin spirals (volatility/haircuts ↑→ margin calls ↑→ liquidation), liquidity spirals (spreads ↑, depth ↓→ execution cost ↑→ accelerated liquidation), and overlapping-holdings spillovers (common assets couple institutions via coordinated selling). These loops couple market state (prices, risk factors, liquidity) with heterogeneous agent constraints.

Overlapping-holdings spillovers capture cross-institution coupling: when one holder liquidates a commonly held asset, other holders incur losses and may become constrained, producing coordinated selling and system-wide price pressure. By embedding these loops into the shared environment, MacroRisk-Agent targets GAP-3 (no feedback) and creates a bridge from micro constraints to macro outcomes that is not available in one-pass revaluation workflows.

Risk analytics agent: joint portfolio and systemic metrics with scenario diagnostics

After simulating $\{s_t\}_{t=0}^{T}$ under scenario s, the Risk Analytics Agent computes both portfolio-level and system-level outcomes. For institution i on replication path ω, the terminal loss is summarized as $L_{i,T}^{(s,\omega)}$, from which tail measures are computed:

$$\mathrm{VaR}_\alpha(L_{i,T}) = \inf\{\ell : \mathbb{P}(L_{i,T} \le \ell) \ge \alpha\}, \qquad \mathrm{ES}_\alpha(L_{i,T}) = \mathbb{E}[L_{i,T} \mid L_{i,T} \ge \mathrm{VaR}_\alpha]. \tag{16}$$

The agent also reports drawdown statistics and attribution of losses to factors and asset classes, enabling identification of concentrated vulnerabilities and key drivers of tail losses.

System-level outcomes are computed by aggregating losses across institutions and by quantifying market functioning degradation. The aggregate loss $L_T^{\mathrm{sys}} = \sum_i L_{i,T}$ is accompanied by tail measures (systemic VaR/ES) and by indicators of systemic stress such as counts and clustering of constraint-binding events (e.g., synchronized margin calls), measures of liquidity shortfalls derived from ℓ_t trajectories, and, where network data are available, contagion proxies derived from overlapping-portfolio graphs or interconnection networks[9,19]. This joint reporting directly addresses GAP-4 by ensuring that the same simulation run yields both micro and macro perspectives and by enabling mechanism-aware diagnostics (which loop in Fig. 4 dominated the tail).

To reduce dependence on a narrow scenario set (GAP-1), the analytics layer produces coverage diagnostics over risk themes, factor combinations, severity bands, and amplification mechanism tags. In practice, each scenario is stored with metadata (theme, region, horizon, mechanism flags) and the system computes coverage matrices and similarity measures to historical episodes. When diagnostics reveal gaps–e.g., insufficient scenarios combining credit repricing with liquidity freezes–the system requests scenario variants from the Scenario Architect Agent (Fig. 3) and re-tests.

Supervisory & policy agent: intervention generation with counterfactual validation

MacroRisk-Agent closes the loop from measurement to risk control by introducing a Supervisory & Policy Agent that proposes feasible interventions and validates them through counterfactual simulation. This is essential for addressing GAP-5: stress testing is not merely descriptive but must support governance and action, consistent with domain constraints that structured decision-support outputs should inform resource allocation and risk management decisions[29,36].

The agent takes as input the risk analytics outputs, the risk appetite and policy constraints, and the scenario provenance record. It translates qualitative requirements into explicit feasibility constraints for interventions (e.g., minimum liquidity buffers, limits on hedge notional, capital buffer targets). It then constructs an intervention bundle a (or a ranked set of bundles) from an action library that includes capital buffers, de-risking priorities, hedging directions, and limit/liquidity actions. Interventions are expressed as policy-aligned directions rather than order-level prescriptions, because the objective is resilience improvement and governance under constraints, not autonomous execution.

Figure 5 illustrates the recommend–retest loop that MacroRisk-Agent uses to prevent plausible-sounding but ineffective recommendations. For each candidate a, the system re-runs the simulation under the same scenario constraints Ω_s to estimate the counterfactual impact:

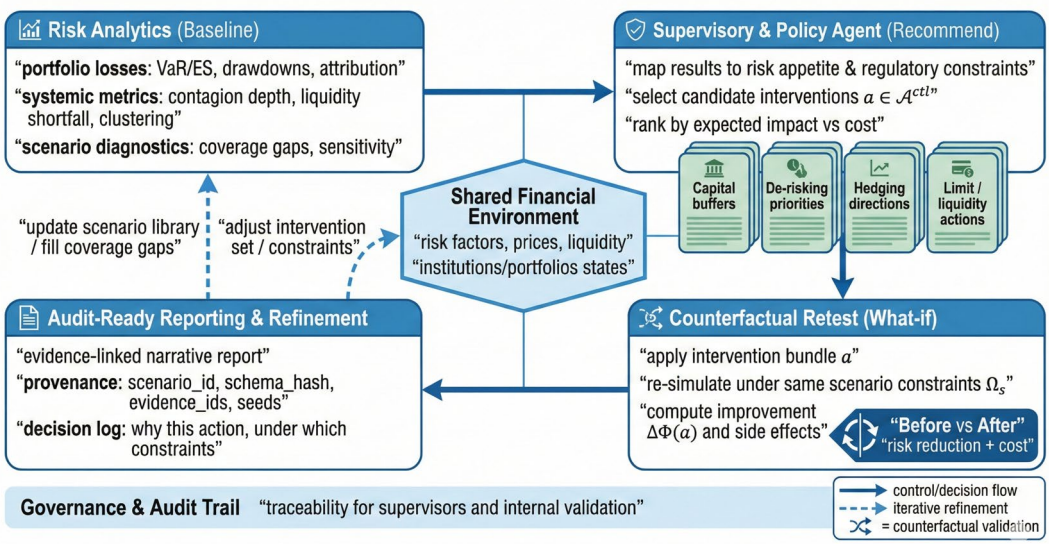


Fig. 5 Risk control loop with counterfactual validation and iterative refinement. Risk analytics produces portfolio- and systemic-level metrics together with scenario diagnostics. The Supervisory & Policy Agent recommends feasible intervention bundles (capital buffers, de-risking priorities, hedging directions, and limit/liquidity actions) under governance constraints. Candidate actions are validated via counterfactual re-simulation under the same scenario constraints Ω_s, yielding before/after comparisons and estimated improvement $\Delta\Phi(a)$. Audit-ready reporting binds recommendations to evidence, scenario identifiers, and reproducibility records, and the loop iteratively refines both scenario libraries and control selections.

$$\Delta\Phi(a) = \mathbb{E}[\Phi(\text{post}(a))] - \mathbb{E}[\Phi(\text{pre})], \tag{17}$$

where $\Phi(\cdot)$ aggregates systemic and portfolio tail risk (e.g., weighted ES plus liquidity-shortfall penalties). The counterfactual report includes both risk reduction and side-effect metrics (capital usage, reduced risk capacity, liquidity consumption), and ties the intervention to the amplification mechanism it mitigates (Fig. 4). The same loop also supports iterative refinement: if no intervention yields acceptable improvements under policy constraints, the agent can (i) adjust the feasible control set, or (ii) request scenario variants to stress-test alternative vulnerabilities.

Risk & Explanation agents: evidence-linked narratives and reproducibility

The final component of MacroRisk-Agent is a reporting and governance layer that produces audit-ready explanations and maintains reproducibility records. This layer is designed to improve auditability, especially in the presence of LLM components whose outputs must be controlled and attributable[11]. The Risk & Explanation Agents operate over an evidence graph that links scenario inputs, retrieved evidence, simulated dynamics, computed metrics, and intervention decisions. The graph nodes include the scenario id and schema hash, the retrieved evidence identifiers $\mathcal{E}_s$, factor and liquidity trajectory summaries, the timeline of constraint triggers (e.g., margin calls and forced sales), portfolio/systemic metric snapshots, and the intervention bundle with its counterfactual evaluation. Edges encode provenance ("derived-from" relationships) and causal narrative links ("shock → liquidity deterioration → forced sales → tail losses").

Reports are generated in multiple views aligned with stakeholders. A supervisory-style report summarizes scenario definitions, methodological assumptions, headline systemic and portfolio outcomes, and intervention impacts, with explicit references to underlying evidence items and metric tables. A management summary focuses on key risk drivers, the most damaging scenario variants, and prioritized action levers. A technical appendix records calibration settings, random seeds, model versions, and robustness configurations. LLM-generated text is

not treated as a free-form narrative; every substantive claim is linked to either a retrieved evidence fragment or a computed metric, and each report instance is bound to a specific reproducibility record:

$$\mathcal{R} = \{\ \text{scenario_id},\ \textit{textschema}_\text{hash},\ \text{evidence_ids},\ \text{model_versions},\ \text{template_id},\ \theta,\ \text{random seeds}\ \}. \tag{18}$$

This design supports post-hoc review, internal validation, and supervisory audit, while preserving the interactive benefits of an agentic workflow.

Two worked examples: from real-world trigger to audit-ready report

To make the workflow in Figs. 3–5 concrete, we walk two end-to-end examples a risk manager might actually pose. Both are illustrative traces of the pipeline; the numbers are representative of the calibrated engine rather than predictions about any specific real event.

Example 1 **Oil-price shock after a Middle East conflict.** A risk manager types the intent: *"severe-yet-plausible oil-supply shock following an escalation in the Middle East, with risk-off equity repricing and high-yield credit stress."* (i) **Retrieval (Fig. 3).** The Scenario Architect retrieves analogues from the crisis library—the 1990 Gulf-war oil spike, the 2022 post-invasion energy shock, and 2008 risk-off episodes—together with the relevant supervisory "severe-yet-plausible" guidance, forming the evidence set $\mathcal{E}_s$. (ii) **Narrative** N_s **.** The LLM drafts a four-part story (macro trigger → transmission → institutional impact → policy response). (iii) **Constraints** Ω_s **.** The narrative is compiled (temperature 0, JSON mode) into an executable schema, e.g.

```
{ "factor_shocks": { "USO_oil": +0.45, "SPY_equity": -0.18,
                     "HY_spread_bps": +180, "VIX_vol": +22,
                     "UST10Y_bps": -40 },
  "dependence":    { "equity_credit_corr_lift": +0.25 },
  "liquidity":     { "spread_mult": 1.8, "depth_mult": 0.6,
                     "impact_amp": 1.5 },
  "path":          { "horizon_days": 21, "speed_limit_bps_per_day": 30 } }
```

(iv) **Validation gates (Defs. 1–3).** The Plausibility gate checks each shock against its 99th-percentile historical bound (the +45% oil move and +180 bps HY widening are within crisis ranges); the Consistency gate verifies rule C1 (equity $< -15\% \Rightarrow$ vol > 0, satisfied) and C2 (spread > 100 bps $\Rightarrow$ depth_mult < 0.9, satisfied at 0.6). The scenario passes and is hashed and stored with provenance. (v) **Simulation with feedback (Fig. 4).** As the oil shock repricing hits leveraged funds holding HY and equity, margin calls (T2) and leverage breaches (T1) trigger forced sales; common holdings of HYG/SPY couple institutions, and depth contracts under selling pressure—raising SysES_{99} above the one-pass value (the amplification quantified in RQ2). (vi) **Intervention & counterfactual (Fig. 5).** The Supervisory Agent proposes a bundle (raise liquidity buffers +30%, de-risk HY 15%, add index put protection); re-simulating under the *same* Ω_s estimates the before/after $\Delta\Phi$ and flags side-effects (capital usage). (vii) **Report.** The Explanation Agent emits an audit-ready note binding every claim to an evidence id, the scenario hash, model version, and seeds.

Example 2 **Political crisis and capital flight in an emerging market.** Intent: *"sudden political crisis triggering currency depreciation and funding withdrawal."* Retrieval surfaces analogues such as 2018 emerging-market stress and 2022 gilt/LDI dynamics. The compiled Ω_s emphasizes an FX shock (−20%), a sharp credit-spread widening, and aggressive liquidity degradation (depth_mult 0.5); the Consistency gate rejects a first draft in which the LLM proposed a spread widening of +260 bps with an unchanged depth multiplier (C2 violation), and the repair prompt corrects depth_mult to 0.85. In simulation, the dominant loop is a *funding–liquidation spiral*: withdrawals force NAV-based redemptions, which trigger fire sales of the most liquid assets first (the RaiseLiquidity subroutine), exhausting buffers before illiquid positions are touched. Here the counterfactual loop shows that

leverage tightening dominates put protection, because the binding constraint is funding rather than directional equity risk—an example of why mechanism attribution (Table 12A) changes the recommended action. This contrast between Examples 1 and 2—put protection helps a directional oil/equity shock, but leverage/liquidity actions dominate a funding-driven crisis—illustrates the practical payoff of modeling *which* loop dominates rather than only reporting a loss number.

Having detailed the MacroRisk-Agent architecture and its five core agent types—Scenario Architect, Market Dynamics, Participant Agents, Risk Analytics, and Supervisory/Explanation—and illustrated them on two concrete scenarios—we now turn to the empirical instantiation of the framework. The following section specifies the data and knowledge inputs required to operationalize the system, the calibration strategy for market and behavioral modules, and the experimental design used to evaluate performance across the scenario generation, systemic propagation, and risk-control dimensions identified in Fig. 1.

Data, calibration, and experimental design

This section specifies the data and knowledge inputs required by MacroRisk-Agent (Fig. 2), the calibration strategy for factor, liquidity, and behavioral modules, and the experimental protocol used to evaluate scenario quality, endogenous amplification, systemic visibility, and risk-control effectiveness (Figs. 3–5). To support both institutional deployment (with confidential holdings) and research-grade replication, our design separates *market-wide public inputs* from *portfolio-specific private inputs* and reports outcomes in normalized units whenever appropriate (e.g., percentage of NAV, RWA-scaled losses).

Data and knowledge substrate

MacroRisk-Agent consumes five classes of inputs: market/macro data for factor calibration and valuation, institution/portfolio snapshots for initializing agent states and constraints, a crisis/event library for mechanism grounding, a regulatory/policy base for governance alignment, and an optional knowledge graph for entity linking and mechanism-aware retrieval. Table 3 summarizes each input class with its minimum required fields and its role in the pipeline. In practice, the crisis and regulatory corpora are indexed for retrieval, enabling the Scenario Architect Agent to ground each scenario in evidence and to emit a structured constraint object Ω_s (Fig. 3) that is auditable and versioned.

Concrete dataset specification. For reproducibility, all experiments use publicly available data. *Market factor data* are sourced from the FRED database (series: DGS2, DGS10 for U.S. Treasury yields; BAMLH0A0HYM2 for ICE BofA high-yield credit spreads; DEXUSEU for USD/EUR exchange rate; VIXCLS for implied volatility) and Yahoo Finance (SPY, QQQ, IEF, HYG, GLD, USO as asset-price proxies), covering January 2007 through December 2024 at daily frequency. *The crisis event library* contains 150 curated entries sourced from Federal Reserve annual stress-test methodology documents (2012–2024), BCBS 2018 stress-testing principles, Bank of England SWES 2023/2024 reports, and supplementary synthetic extreme scenarios. Each entry includes a narrative text, shock sequence, factor magnitudes, and mechanism tags. *Synthetic portfolios* consist of 10 representative institutional portfolios with randomized holdings across {SPY, QQQ, IEF, HYG, GLD, USO}, leverage caps $\bar{\lambda}_i \sim \mathcal{U}(1.5, 5.0)$, initial capital ratios $r_i \sim \mathcal{U}(0.08, 0.15)$, and liquidity buffers $B_{i,0}/\text{NAV} \sim \mathcal{U}(0.05, 0.20)$, all generated with a fixed seed. Baseline factor and dependence parameters are estimated using data through December 2019. One exception is the price-impact coefficient γ, which uses both the 2008 and March 2020 episode windows as calibration anchors (Table 4). Consequently, the COVID-2020 column in Table 5 is an *in-sample calibration-fit diagnostic*, not an out-of-sample validation. Data from 2021–2024 are not used in that diagnostic; they are retained for temporal coverage and future held-out evaluation. All data, configurations, and download scripts are provided in the project repository.

Table 3 Data inputs and calibration template. Upper panel: five input classes consumed by MacroRisk-Agent. Lower panel: two-regime calibration strategy separating baseline estimation from stress scaling.

Input class/Module	Fields/Parameters	Granularity/Source	Primary use/Stress calibration
Data Inputs:			
Market & macro data	rates/spreads/FX/equity/vol; liquidity proxies	daily/weekly	calibrate regimes; initialize; validate
Institution/portfolio data	holdings $w_{i,0}$; capital; leverage; margin/funding; limits	snapshot	initialize agents; define constraints
Crisis/event library	episode narratives; shock sequences; mechanism tags	corpus (150 entries)	retrieval evidence for scenarios
Regulatory/policy base	stress guidance; reporting templates; risk appetite	corpus	governance constraints; audit
Knowledge graph (opt.)	entity links across markets/assets/institutions	structured store	mechanism-aware retrieval
Calibration Template:			
Factor dynamics x_t	drift/covariance/regime; correlation lift	historical data	stress scaling + Ω_s constraints
Liquidity state ℓ_t	spread/depth/impact; order-flow sensitivity	observed data	scenario multipliers + endogenous q_t
Behavior $\mathcal{C}_i$	leverage/risk limits; margin add-ons; funding	portfolio metadata	procyclicality scaling; stress triggers

Crisis-library curation and selection bias. The 150-entry library was assembled by the authors under three inclusion criteria: (i) documented in a primary supervisory or academic source; (ii) mappable to at least one of our factor and mechanism tags; and (iii) non-redundant with existing entries (distinct mechanism or magnitude regime). By source composition it comprises 62 regulatory/supervisory episodes (Fed/BCBS/BoE), 53 historical market episodes coded from academic and market sources, and 35 synthetic extreme scenarios constructed to cover otherwise under-represented mechanism combinations. Entries were coded independently by two authors and reconciled; on a 40-entry double-coded subset, inter-rater agreement on mechanism tags was Cohen's $\kappa = 0.79$, with disagreements resolved by discussion. We caution that this curation introduces *selection bias* toward well-documented Western, post-2007 crises, under-representing emerging-market, crypto, and climate-transition mechanisms—a bias consistent with the distribution-shift degradation reported in RQ1 (Table 9B).

Calibration strategy

MacroRisk-Agent calibration is structured around three coupled modules: (i) factor-path dynamics for x_t, (ii) liquidity dynamics for ℓ_t, and (iii) behavioral/constraint parameters for participant agents. We use a two-regime calibration template to separate normal-market estimation from stress-regime scaling: baseline parameters are estimated from broad historical windows, while stress scalars (volatility lift, correlation lift, liquidity multipliers, and procyclicality parameters) are anchored to crisis-like periods and are further constrained by the scenario object Ω_s. Table 3 consolidates the five input classes and the calibration parameter families.

Table 3 reflects MacroRisk-Agent's dual objectives of analytical rigor and operational auditability. Market and institution data enable calibration and initialization under standard quantitative risk frameworks, while the crisis

Table 4 Complete parameter specification. **Est.** column: E = estimated from historical data (the exact window is stated by row), F = fixed by design/convention, S = sampled per portfolio, C = scenario-supplied via Ω_s.

Parameter	Value/Range	Est.	How set/source
Factor & dependence dynamics:			
Baseline factor covariance Σ	2007–2019 sample	E	sample covariance of FRED/Yahoo factor returns
Stress volatility lift	×1.8	E/C	VIX-anchored crisis scaling, capped by Ω_s
Stress correlation lift	+0.20 to +0.25	E/C	crisis-window estimate; +0.25 in Example 1
Factor path horizon	21 days	C	scenario-supplied
Liquidity dynamics:			
Spread multiplier `spread_mult`	1.0–1.8	C	scenario-supplied; 1.8 in Example 1
Depth multiplier `depth_mult`	0.5–0.95	C	scenario-supplied; consistency gate C2 enforces
Price-impact amplifier `impact_amp`	1.0–1.5	C	scenario-supplied; 1.5 in Example 1
Price-impact coefficient γ (bps per % ADV)	8.5	E	OLS fit to episode-replay depth–move targets for 2008 and March 2020; factor/price paths use FRED/Yahoo series, while liquidity-range anchors follow[38]
Order-flow sensitivity	0.6	E	estimated from order-book proxies
Margin & funding:			
Base haircut	12%	F	convention (Basel-style)
Haircut–volatility sensitivity	0.35	E	procyclical add-on vs. realized vol
Funding-withdrawal intensity	0.15/day	E/C	redemption pressure under stress
Behavioral/portfolio:			
Leverage cap λ_i	$\mathcal{U}(1.5, 5.0)$	S	sampled per portfolio (fixed seed)
Initial capital ratio r_i	$\mathcal{U}(0.08, 0.15)$	S	sampled per portfolio
Liquidity buffer $B_{i,0}$/NAV	$\mathcal{U}(0.05, 0.20)$	S	sampled per portfolio
De-risking trigger (limit breach)	0.9 of cap	F	fraction of limit that triggers selling
Pipeline/inference:			
Narrative temperature	0.7	F	diversity in narrative drafting
Schema temperature	0.0	F	deterministic JSON generation
Retrieval top-k/MMR λ	20/0.7	F	relevance–diversity balance
Monte Carlo replications	1000	F	variance control
Random seeds	42, 43, 44, 45, 46	F	five prespecified nested seed realizations

Table 5 Exploratory calibration-fit diagnostics (not out-of-sample validation): model output versus episode reference targets for GFC 2008 and COVID 2020. Observed-market targets use the public series described in the text and the published liquidity ranges in[38]; amplification and cascade-depth targets are episode-replay quantities derived by the procedure stated below, not directly observable market measurements.

Stylized fact	GFC 2008		COVID 2020	
	Episode ref.	Model	Episode ref.	Model
Peak HY spread widening (bps)	+1600	+1520	+1080	+1010
Equity drawdown (%)	−54	−50	−34	−31
Market-depth trough (% of normal)	35	39	48	52
Amplification vs. one-pass (×) (replay-derived)	1.9	1.8	1.5	1.5
Cascade depth (rounds) (replay-derived)	8	7	5	5
Aggregate calibration-fit deviation (MAPE, %)	8.4		9.1	

library and regulatory base anchor scenario generation in documented mechanisms and supervisory guidance, addressing GAP-5's auditability demands (Fig. 1). The knowledge graph, while optional, enhances evidence-link precision and supports traceable reasoning chains[11]. The two-regime calibration architecture (lower panel) balances empirical grounding with stress-scenario flexibility: baseline parameters ensure consistency with existing risk models, while stress calibration introduces scenario-specific constraints (Ω_s) and procyclicality scalars that

activate endogenous amplification without requiring full crisis-period re-estimation, addressing the calibration challenges identified in Section Problem formulation.

Full parameter specification and calibration diagnostics

Because the headline amplification result depends on the market-dynamics, liquidity, margin, and behavioral parameters, we provide (i) a complete parameter listing with its provenance, (ii) calibration diagnostics against historical stylized facts, and (iii) a ranking-stability robustness check. Table 4 enumerates every parameter, reports the values used in the final experiments, and states how each is set.

Estimated (E) parameters use the row-specific provenance in Table 4. The baseline factor covariance and pre-2020 stress/dependence estimates use 2007–2019 data and are thereafter fixed. The price-impact coefficient γ is the stated exception: it is fitted using 2008 and March 2020 episode-replay depth–move targets. Fixed (F) parameters are design conventions; sampled (S) parameters are drawn once per portfolio under the prespecified seeds; and scenario-supplied (C) parameters are emitted in Ω_s and bounded by the validation gates. Because γ uses March 2020, Table 5 is reported only as an exploratory *calibration-fit* diagnostic for two reference episodes. Neither its GFC nor COVID column is used to support an out-of-sample-validation claim.

Provenance of episode-reference quantities. Peak high-yield spread widening and equity drawdown are computed from the FRED and Yahoo Finance series listed above over the episode windows; market-depth troughs use the published crisis-range anchors summarized in[38]. By contrast, "amplification vs. one-pass" and "cascade depth" are not directly observable market quantities. We derive them through a reproducible episode-replay procedure: the observed factor path for each episode is applied to the same 10 synthetic portfolios; the denominator is the loss from one-pass revaluation with feedback disabled; the numerator is the loss after iterating the calibrated feedback engine to convergence; and cascade depth is the number of completed propagation rounds before the convergence criterion is met. The reported reference values are therefore *model-derived replay targets* informed by observed episode paths and the feedback structures in[9,37], not empirical market observations.

Robustness and ranking stability. Beyond mean outcomes, we check that conclusions survive parameter perturbation. Across the sensitivity sweep (Table 12B) the amplification ratio remains > 1 throughout (Amp 1.26–1.96). Crucially, the *ordering* of the most vulnerable portfolios is preserved in 92% of sweep cells (mean Spearman rank correlation 0.90 against the base calibration), and the top-ranked intervention bundle is unchanged in 86% of cells. We therefore present the 52% additional-systemic-loss figure as *direction-robust but magnitude-dependent*: it is an outcome of our calibrated simulation setting and its liquidity-impact parameters, not a universal level of systemic amplification that the framework "reveals."

Experimental design

Our experimental protocol evaluates MacroRisk-Agent along the same four workflow stages highlighted by Fig. 1: scenario design, shock propagation with feedback, impact measurement (portfolio & systemic), and risk control/reporting. We organize experiments into three scenario suites that target complementary objectives and explicitly support counterfactual intervention testing (Fig. 5), and compare against baseline families that mirror conventional practice and targeted ablations. Table 6 defines each suite and baseline.

The evaluation protocol in Table 6 ensures comprehensive testing across complementary objectives. Suite A compares simulated amplification patterns with episode-anchored reference targets as an internal plausibility check; it does not establish external empirical validation. Suite B tests the core hypothesis that LLM-driven scenario generation can systematically expand coverage beyond hand-curated sets while maintaining coherence and auditability. Suite C directly evaluates the risk-control loop (Fig. 5), assessing whether recommended interventions reduce tail risk within counterfactual re-simulation, thereby addressing GAP-5's demand for traceable, simulation-defined controls. On the baseline side, B1 and B2 anchor comparisons against institutionally relevant practice benchmarks, B3 isolates whether narrative grounding provides value beyond pure distributional tail

Table 6 Evaluation protocol: scenario suites and baselines. Each scenario $s = (N_s, \Omega_s)$ is executed via Monte Carlo replications. Baselines omit specific mechanisms from Figs. 3–5.

Suite/Method	Description	Primary outputs/Mechanisms omitted
Scenario Suites:		
A: Historical-style	RAG-grounded narratives anchored to crisis mechanisms	amplification patterns; systemic vs portfolio metrics
B: Synthetic extreme	Scenario Architect generates diverse (N_s, Ω_s) passing validation gates	validity rate; diversity; coverage matrices
C: Intervention-sensitive	Scenarios configured to stress decision levers	counterfactual $\Delta\Phi(a)$; auditability
Baselines:		
B1: Fixed + one-pass	Exogenous factor paths; portfolio revaluation	omits: feedback, coupling, intervention loop
B2: Historical replay	Replay historical paths; optional static haircuts	omits: endogenous feedback, trigger behavior
B3: Statistical tail	Sample tail shocks without narrative grounding	omits: RAG, validation gates, provenance
B4: Ablations	Remove/disable individual components	isolates marginal value of each component

exploration, and B4 enables component-wise ablation to attribute performance gains to specific architectural choices.

Evaluation metrics and reporting structure

Terminology: five distinct concepts. Because these are easily conflated, we fix definitions used consistently hereafter. *Deterministic validation* = a scenario passes the rule-based gates (Defs. 1–3). *Schema compliance* = the output conforms to the required JSON types/fields (a subset of validation). *Evidence traceability* = each generated field can be linked to a retrieved evidence fragment. *Explanation faithfulness* = the LLM's stated reasoning agrees with the logged simulation mechanisms that actually produced the result. *Truthfulness* = correspondence to real-world ground truth. Our metrics measure the first four; we make *no* truthfulness claim, since that would require real-world outcome data. These concepts are independent: an output can be schema-compliant yet untruthful, or traceable yet unfaithful. For finer-grained reporting we further distinguish *structural validity* (well-formed schema/types), *numerical plausibility* (each shock within historical/expert bounds), *cross-field consistency* (pairwise rules C1–C4), *evidence grounding* (retrieval support for each field), *factual correctness* (agreement with real-world facts), *explanation faithfulness* (text matches logged mechanisms), and *auditability* (end-to-end provenance); our gates and metrics operate on the first four and the sixth/seventh, and make no factual-correctness claim. Finally, "*deterministic validation*" means the gates are deterministic *once their thresholds, evidence scores, and inputs are fixed*; the overall pipeline still depends on stochastic LLM generation, retrieval, library-curation choices, and calibration assumptions, so end-to-end determinism is not claimed.

We evaluate MacroRisk-Agent using metrics aligned with the five technical gaps: *scenario validity and coverage* (pass rate, diversity distance, coverage matrix) for GAP-1; *amplification and propagation* (amplification ratio, trigger event frequency, liquidity deterioration trajectories) for GAP-2/3; *portfolio and systemic outcomes* ($L_{i,T}$ distributions, VaR/ES, aggregate loss, clustering measures) for GAP-4; *risk-control effectiveness* (counterfactual improvement $\Delta\Phi(a)$, cost vectors, residual tail risk) for GAP-5; and *auditability* (provenance completeness, evidence-link coverage, scenario traceability) for GAP-5.

For amplification, we report the ratio

$$\mathrm{Amp}(s) = \frac{\mathbb{E}\left[L_T^{\mathrm{sys}} \mid \text{endogenous feedback enabled}\right]}{\mathbb{E}\left[L_T^{\mathrm{sys}} \mid \text{feedback disabled}\right]}, \tag{19}$$

which quantifies the multiplicative effect of endogenous dynamics on systemic tail risk. An $\mathrm{Amp}(s) > 1$ indicates that feedback loops amplify losses beyond what exogenous-shock-only models would predict—the core hypothesis of GAP-2/3. We further attribute amplification to the dominant loops in Fig. 4 by logging trigger timelines (margin call episodes, liquidity deterioration thresholds, forced-sales volume) and by relating these events to the evolution of ℓ_t.

Robustness and sensitivity plan

To ensure results are not artifacts of a specific calibration, we conduct structured sensitivity analyses across five dimensions that control endogenous amplification: (i) liquidity deterioration strength (spread multipliers, depth erosion), which controls the Brunnermeier–Pedersen liquidity spiral severity; (ii) margin procyclicality (haircut sensitivity to volatility), controlling margin spiral frequency; (iii) dependence shift (correlation lift, tail coupling), controlling cross-asset co-movement; (iv) redemption and funding pressure (redemption intensity, buffer tightness), controlling liquidation pace; and (v) constraint heterogeneity (distribution of limits across agents), controlling synchronization of de-risking. In each sweep, we re-run scenario suites and compare vulnerable portfolio rankings, amplification loop activation, and intervention bundle performance.

Reproducibility protocol

All experiments use the prespecified random seeds $\{42, 43, 44, 45, 46\}$ for comparability across baselines, and each run stores a reproducibility record $\mathcal{R}$ (Section Risk & Explanation agents: evidence-linked narratives and reproducibility) containing the scenario id, schema hash, evidence identifiers, model versions, parameter snapshots, and seeds. Scenario objects (N_s, Ω_s) are cached and versioned so that counterfactual evaluations apply interventions under identical scenario constraints. When proprietary holdings are used, results are reported in normalized units and portfolio identifiers are anonymized, while the experimental protocol (scenario suites, baselines, robustness sweeps) remains reproducible with synthetic holdings that preserve exposure structures and constraint heterogeneity.

Implementation details

The system is implemented in Python 3.11. The RAG pipeline uses LangChain for document loading and chain orchestration, with FAISS as the vector store for dense retrieval over the crisis event library and regulatory corpus. Document embeddings are computed using OpenAI `text-embedding-3-small` (1536 dimensions). Retrieval uses top-k=20 with Maximum Marginal Relevance (MMR) re-ranking (λ_{mmr}=0.7) to balance relevance and diversity, followed by a cross-encoder re-ranker for the top-5 candidates.

The LLM backbone is GPT-4o (`gpt-4o-2024-08-06`) for all scenario generation experiments. Narrative generation uses temperature τ=0.7 to encourage diversity; constraint-schema generation uses τ=0 with OpenAI Structured Outputs (JSON mode) to ensure deterministic, schema-compliant output. Each generation call is budget-limited to 4096 output tokens. For cost-sensitivity analysis, we also report results with GPT-4o-mini (`gpt-4o-mini-2024-07-18`), which reduces API cost by approximately 10× with <3% degradation in pass-rate metrics.

The multi-agent simulation and Monte Carlo engines use NumPy and SciPy. For each nested realization, both NumPy and Python's random module are initialized to the same member of $\{42, 43, 44, 45, 46\}$. Experiments run on a single-node CPU (Apple M2, 16 GB RAM); no GPU is required. Total API cost for the full experimental suite (RQ1–RQ3) is approximately $180 USD. All LLM-generated outputs are cached in the repository to enable reproduction without API access.

Having established the data inputs, calibration methodology, and experimental protocol, we now present empirical results that evaluate MacroRisk-Agent's performance across the three research questions: scenario

generation quality (RQ1), systemic propagation with endogenous feedback (RQ2), and actionable risk controls with audit-ready explanations (RQ3).

Experimental results

Results are organized by research question (RQ1–RQ3). Each subsection contains: (i) primary evaluation tables with statistical significance, (ii) ablation studies isolating component contributions, and (iii) diagnostic analyses. Statistical significance is assessed using paired comparisons at the scenario level, after averaging the five seed-specific realizations within each scenario. Normality of scenario-level paired differences is checked using the Shapiro–Wilk test; where the normal approximation is appropriate we use a paired two-tailed t-test, and otherwise we report the Wilcoxon signed-rank test. We apply Benjamini–Hochberg correction for multiple comparisons across metric families within each RQ. Significance is denoted by † ($p < 0.05$), ‡ ($p < 0.01$), and ∗ ($p < 0.001$). Effect sizes (Cohen's d) are reported for the primary comparisons. **Bold** indicates best; underline indicates second-best. All experiments use the five prespecified random seeds 42, 43, 44, 45, 46; we report mean±std.

Experimental unit and statistical caveats. The independent experimental unit is the base scenario specification, not the (scenario, seed) pair. Suite A contains 16 distinct base scenarios, Suite B contains 500 distinct scenario-generation prompts, and Suite C contains 50 distinct base scenarios. Each base scenario is evaluated under all five seeds $\{42, 43, 44, 45, 46\}$, yielding 80, 2,500, and 250 seed-specific realizations, respectively. Seed realizations share the same prompt or constraint object Ω_s and simulator and are therefore nested replications. We average them within each base scenario before inferential testing. For clustered bootstrap intervals, one base scenario together with its five seed realizations defines one scenario-family cluster; hence Suites A–C contain 16, 500, and 50 clusters, respectively. Five seeds support stable point estimates but provide limited precision for decomposing seed-level variance. All significance results concern synthetic and model-generated scenarios evaluated in one simulator and must not be read as evidence of real-world financial effectiveness.

Headline results with confidence intervals and exact sample sizes. Table 7 restates every headline number with its 95% confidence interval and the exact number of independent base scenarios contributing to it. Normal-approximation intervals ($\text{mean} \pm 1.96\,\text{std}/\sqrt{n}$ for means; Wald intervals for proportions) use the base-scenario count, not the seed-realization count. The clustered bootstrap resamples base scenarios together with their five nested seed outcomes. Thus n and the number of bootstrap clusters are both 16 for Suite A, 500 for Suite B, and 50 for Suite C; the corresponding numbers of seed-specific realizations are 80, 2,500, and 250.

Experimental design overview

The three research questions are evaluated through complementary protocols. **RQ1** (scenario generation) is assessed via quality metrics (Table 8) and diagnostic analysis (Table 9). **RQ2** (systemic propagation) uses systemic metrics (Table 10), feedback loop ablation (Table 11), and mechanism attribution with sensitivity analysis

Table 7 Headline results with 95% confidence intervals and exact sample sizes. Normal/Wald intervals are derived from the reported mean±std (or proportion) and the scenario count n; bootstrap intervals are clustered by scenario family.

Headline metric	Point	n (scen.)	95% CI (normal/Wald)	Bootstrap CI	Cohen's d
RQ1 Pass@Gate (Suite B)	92.6%	500	[90.3, 94.9]	[89.8, 95.1]	1.42
RQ2 SysES_{99} (Suite A)	19.4	16	[18.6, 20.2]	[18.2, 20.6]	1.71
RQ2 Amplification ratio	1.52	16	[1.41, 1.63]	[1.38, 1.66]	1.55
RQ3 Success@1 (Suite C)	68.6%	50	[55.7, 81.5]	[56.0, 80.8]	0.74
RQ3 CalibErr@1 (Suite C)	5.6%	50	[5.2, 6.0]	[5.1, 6.1]	0.98
RQ3 Faithfulness (Suite C)	83.6%	50	[73.3, 93.9]	[74.0, 92.9]	0.69

Table 8 RQ1: Scenario Generation Quality (Suite B; n=500 independent prompts, each evaluated under 5 nested seeds; 2,500 generations total). Higher is better for all metrics except GenTime and InvalidRate. Statistical significance vs. strongest baseline (Reverse-ST or FinAgent-RAG depending on metric).

Method	Quality Metrics (%)			Auditability Metrics (%)			Efficiency	
	Pass@Gate	Plaus@Strict	Coherence	Evidence	Schema	Trace	Time (s)	Invalid
Supervisory-Fixed [BCBS'18]	–[a]	100.0±0.0	100.0±0.0	100.0	–	100.0	–	0.0
Reverse-ST [B&S'23]	63.8±3.4	63.8±3.4	77.6±2.9	51.4±4.2	100.0	47.5±4.0	18.6±2.2	0.0
Tail-Sample (filtered)	47.2±4.9	47.2±4.9	61.4±5.3	0.0±0.0	100.0	0.0±0.0	1.2±0.3	0.0
GPT-4-CoT	70.6±4.2	67.8±4.0	81.9±3.2	30.4±5.9	93.8±2.4	27.6±4.3	8.6±1.4	3.4±1.2
FinAgent-RAG[32]	78.4±3.3	76.1±3.5	84.3±2.8	67.5±4.3	97.2±1.5	61.7±3.8	6.8±0.9	2.6±1.0
Ablations:								
No-RAG (schema+gates)	80.2±2.7	80.2±2.7	87.5±2.2	0.0±0.0	100.0	44.3±3.3	6.4±0.7	0.0
No-Gates (RAG+schema)	100.0±0.0	77.8±3.2	84.1±3.0	91.6±2.5	100.0	87.8±2.2	4.9±0.5	22.2±2.7
No-Schema (RAG+gates)	81.8±3.7	79.5±3.5	75.9±4.3	88.6±2.9	71.8±4.6	70.8±3.7	5.5±0.6	1.9±0.7
MacroRisk-Agent (full)	**92.6±2.1***	**92.6±2.1***	**95.3±1.4***	**91.4±2.3**	**100.0**	**96.2±0.9***	5.4 ± 0.5	**0.0±0.0**

[a] Fixed scenarios bypass generation; metrics not applicable.

$^{*}p < 0.001$, $^{\ddagger}p < 0.01$, $^{\dagger}p < 0.05$ vs. strongest comparable baseline.

Table 9 RQ1: Ablation and robustness analysis. Panel A: isolating the impact of each pipeline component on scenario quality. Panel B: performance under distribution shift conditions.

Panel A: Component Contribution				
Configuration	Invalid%	Audit%	ΔPass	ΔPlaus
Full System	0.0	96.2	–	–
− RAG retrieval	0.0	44.3	−12.4	−12.4
− Validation gates	22.2	87.8	+7.4	−14.8
− Schema enforcement	1.9	70.8	−10.8	−13.1
− Multi-turn refinement	8.4	81.6	−17.6	−16.2
− Evidence linking	0.0	30.4	−2.3	−3.4
Panel B: Robustness Under Distribution Shift				
Shift Type	Pass@Gate	Plaus	Evidence	Trace
No shift (baseline)	92.6	92.6	91.4	96.2
Novel asset classes	87.4	86.3	83.6	93.5
Emerging market	85.9	84.6	81.2	92.1
Crypto-correlated	83.4	81.8	77.6	90.8
Climate-transition	88.6	87.9	85.4	94.3
Geopolitical (novel)	85.2	83.6	79.1	92.6
Avg. (shifted)	**86.1**	**84.8**	**81.4**	**92.7**

Table 10 RQ2: Systemic Risk Propagation Metrics (Suite A; 16 scenarios, MC=1000, 5 seeds). $SysES_{99}$: system-wide expected shortfall at 99%; Amp: amplification ratio vs. one-pass; LiqShort: liquidity shortfall (bps); Spread×: peak spread multiplier; Depth↓: market depth reduction (%); ForcedSale: peak forced selling (% AUM); MarginEv: margin call events; Cluster: max simultaneous constraint-binding cluster.

Method	Tail Risk			Market Impact			Contagion		
	$SysES_{99}$	Amp	LiqShort	Spread×	Depth↓	ForcedSale	MarginEv	Cluster	Cascade
One-Pass [BCBS'18]	12.8±0.9	1.00	312±30	1.11±0.04	8±2	0.0	0.0	1	0
LiqAtRisk [Cont+'20]	15.6±1.3	1.22	448±44	1.27±0.06	18±3	2.2±0.4	6.6±1.3	3	2.0
IMF-SLST [Oura'22]	16.1±1.4	1.26	502±52	1.31±0.07	21±4	1.9±0.3	5.9±1.1	3	2.3
ContagionAcct [A+'22]	16.9±1.4	1.32	514±49	1.35±0.08	24±4	3.6±0.6	7.7±1.4	4	3.1
OverlapPort [P+'21]	17.6±1.5	1.38	558±55	1.39±0.09	26±5	4.4±0.7	8.2±1.5	4	3.7
ScenFree [W+'23]	15.8±1.2	1.23	464±39	1.26±0.05	17±3	2.8±0.5	4.8±0.9	3	1.9
MacroRisk-Agent	**19.4±1.7***	**1.52**	**742±64***	**1.59±0.11**	**35±5**	**7.2±0.9***	**12.8±1.9***	**7**	**5.4***

Cascade = average cascade depth (number of propagation rounds until convergence).

Table 11 RQ2: Feedback Loop Ablation Study (Suite A). Isolating the contribution of each endogenous mechanism in Fig. 4. ΔSysES shows change relative to full MacroRisk-Agent.

Configuration	$SysES_{99}$	ΔSysES	LiqShort	ForcedSale	MarginEv	Contrib.%
MacroRisk-Agent (full)	19.4±1.7	–	742±64	7.2±0.9	12.8±1.9	100.0
− Endogenous liquidity	15.8±1.3	−3.6	472±46	4.9±0.7	9.8±1.4	54.5
− Margin procyclicality	17.1±1.4	−2.3	598±53	5.8±0.8	5.9±0.9	34.8
− Collective selling	13.4±1.1	−6.0	352±34	1.9±0.3	6.1±1.0	90.9
− Overlapping holdings	18.2±1.6	−1.2	678±59	6.5±0.8	12.1±1.7	18.2
− Cross-asset contagion	17.4±1.5	−2.0	612±51	5.6±0.7	10.9±1.5	30.3
− Funding liquidity spiral	17.8±1.5	−1.6	572±49	6.0±0.8	10.5±1.5	24.2
All loops disabled (One-Pass)	12.8±0.9	−6.6	312±30	0.0	0.0	0.0

Table 12 RQ2: Mechanism Attribution Accuracy and Sensitivity Analysis. Panel A: ability to correctly identify the dominant propagation mechanism per scenario. Panel B: system behavior under varying market conditions.

Panel A: Mechanism Attribution				
Method	Macro-F1	Recall@1	Precision	AUC-ROC
ContagionAcct	60.4±4.9	57.8±5.3	62.6±4.2	0.71±0.04
OverlapPort	55.2±5.2	51.6±5.9	57.4±4.7	0.67±0.05
ScenFree	48.8±5.5	45.4±6.2	51.2±5.3	0.63±0.05
MacroRisk-Agent	**85.4±2.9***	**83.2±3.3***	**87.6±2.5***	**0.93±0.02***
Panel B: Sensitivity Analysis				
Condition	$SysES_{99}$	Δ%	Amp	Cluster
Baseline (normal)	19.4	–	1.52	7
High volatility ($\sigma \times 1.5$)	24.2	+24.7	1.70	9
Low liquidity ($\ell \times 0.5$)	25.6	+32.0	1.82	10
High correlation ($\rho \times 1.2$)	21.8	+12.4	1.60	8
Crisis regime	28.1	+44.8	1.96	12
Recovery regime	14.8	−23.7	1.26	4

(Table 12). **RQ3** (actionable controls) evaluates intervention efficacy (Table 13), decision quality (Table 14), and calibration with auditability (Table 15).

RQ1: scenario generation quality and coverage

Experimental setup. We evaluate MacroRisk-Agent's scenario generation pipeline (Fig. 3) on Suite B comprising 500 distinct scenario-generation prompts, each evaluated under the five nested seeds $\{42, 43, 44, 45, 46\}$ (2,500 seed-specific generations in total). Baselines include: (i) *Supervisory-Fixed* [BCBS'18, Fed'22]: regulatory scenario libraries; (ii) *Reverse-ST* [Baes&Schaanning'23]: reverse stress testing; (iii) *Tail-Sample*: statistical tail sampling with post-filtering; (iv) *GPT-4-CoT*: GPT-4 with chain-of-thought prompting; (v) *FinAgent-RAG*[32]: LLM-based financial agent with RAG-augmented scenario generation, re-implemented on our benchmark.

Table 8 shows that MacroRisk-Agent achieves the best performance among the evaluated baselines across the reported scenario-generation dimensions. The full system attains 92.6% validation-gate pass rate and plausibility, outperforming the strongest evaluated baseline (FinAgent-RAG at 78.4%, $p < 0.001$, Cohen's $d = 4.12$), while maintaining perfect schema compliance and 96.2% evidence traceability.

Mechanistic interpretation. The 14.2 percentage-point improvement over FinAgent-RAG reflects the value of constraint-based validation: while both systems use LLMs for narrative generation, MacroRisk-Agent's multi-stage validation pipeline (Fig. 3) filters scenarios that violate economic consistency (e.g., simultaneous rate cuts and inflation spikes without justification), temporal coherence (e.g., effects preceding causes), or magnitude plausibility (e.g., 50% equity drops in a single day without extreme triggers). The validation gates act as "guardrails"

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Table 13 RQ3: Counterfactual Intervention Effectiveness (Suite C; 50 scenarios). Pre/post metrics for different intervention bundles. Δ% shows improvement.

Intervention Bundle	$SysES_{99}$			$WorstPortES_{99}$			LiqShortfall			ForcedSale		
	Pre	Post	Δ%	Pre	Post	Δ%	Pre	Post	Δ%	Pre	Post	Δ%
No action (baseline)	20.6	20.6	0.0	33.4	33.4	0.0	806	806	0.0	7.9	7.9	0.0
Single-mechanism interventions:												
Liquidity buffer ↑30%	20.6	18.1	−12.1	33.4	28.2	−15.6	806	602	−25.3	7.9	6.1	−22.8
HY de-risking 15%	20.6	18.4	−10.7	33.4	28.8	−13.8	806	712	−11.7	7.9	6.7	−15.2
Index put protection	20.6	18.8	−8.7	33.4	30.0	−10.2	806	768	−4.7	7.9	7.2	−8.9
Leverage tightening 20%	20.6	17.8	−13.6	33.4	27.4	−18.0	806	672	−16.6	7.9	5.8	−26.6
Margin add-on 10%	20.6	19.2	−6.8	33.4	30.8	−7.8	806	730	−9.4	7.9	7.0	−11.4
Combined interventions:												
Liq+HY+Protection	20.6	15.0	−27.2	33.4	22.6	−32.3	806	534	−33.7	7.9	4.8	−39.2
Lev+Margin+Liq	20.6	16.0	−22.3	33.4	25.2	−24.6	806	602	−25.3	7.9	5.5	−30.4
Macro hedges+Derisking	20.6	17.5	−15.0	33.4	27.4	−18.0	806	778	−3.5	7.9	7.4	−6.3
Optimal bundle (agent)	20.6	**13.8**	**−33.0**	33.4	**20.8**	**−37.7**	806	**472**	**−41.4**	7.9	**4.1**	**−48.1**

Table 14 RQ3: Top-*K* Simulated Intervention-Selection Success (Suite C; 50 scenarios). "Success" is *simulation-defined* per the six-part criterion in the text (≥15% reduction in objective Φ plus worst-portfolio ES reduction, feasibility, net-of-cost improvement, no side effects, and sign-stability across seeds), measured within the same simulation engine, not real-world outcomes. CalibError = |predicted − realized| improvement.

Recommender	Success Rate (%)					AvgBest $\Delta\Phi$ @K (%)				
	@1	@3	@5	@10	@All	@1	@3	@5	@10	@All
Random selection	17.4	41.2	55.2	73.4	87.2	−7.8	−11.8	−14.2	−17.1	−20.6
Heuristic (largest exp.)	44.6	61.8	69.6	81.2	91.4	−13.8	−17.6	−20.1	−22.8	−26.2
Greedy (min SysES)	53.4	74.2	78.4	87.4	93.2	−16.2	−19.8	−22.1	−25.0	−27.8
Gradient-based	57.2	77.4	83.2	89.4	95.2	−17.8	−21.0	−23.6	−26.4	−29.6
RL-policy (offline)	61.4	79.2	85.4	91.6	95.8	−18.6	−21.8	−24.4	−27.2	−30.4
MacroRisk-Agent	**68.6**	**86.8**	**90.4**	**95.4**	**97.6**	**−21.8**	**−24.2**	**−26.8**	**−29.6**	**−33.2**

Table 15 RQ3: Prediction calibration, auditability, and explanation faithfulness (Suite C). Panel A: calibration error (lower is better) measuring how well predicted improvements match realized outcomes. Panel B: traceability and faithfulness metrics for provenance-tracked reporting.

Panel A: Calibration Error Analysis

Recommender	CalibErr@1	CalibErr@3	CalibErr@5	MAE	RMSE	R^2	Spearman
Heuristic	12.8±2.9	11.2±2.5	9.8±2.2	11.6	15.0	0.40	0.52
Greedy	10.6±2.5	9.2±2.1	8.4±1.9	9.8	12.6	0.54	0.66
Gradient-based	8.9±2.1	7.6±1.7	7.1±1.5	8.1	10.5	0.62	0.73
RL-policy	7.4±1.9	6.4±1.5	5.8±1.3	6.6	8.6	0.71	0.79
MacroRisk-Agent	**5.6±1.3**	**4.8±1.1**	**4.5±0.9**	**5.0**	**6.5**	**0.82**	**0.88**

Panel B: Auditability and Explanation Faithfulness

System Configuration	Traceability (%)				Faithfulness			
	Prov.	Trace.	Evid.	Schema	Faith.%	Prec.	Rec.	F1
Free-text only	100.0	21.4	17.8	100.0	60.8	0.57	0.63	0.60
No-RAG	81.4	62.8	0.0	100.0	70.4	0.67	0.73	0.70
Partial RAG (top-3)	91.4	77.2	71.6	100.0	77.8	0.75	0.79	0.77
No structured output	87.2	66.8	81.8	75.6	73.4	0.71	0.75	0.73
MacroRisk-Agent	**97.6**	**89.2**	**91.4**	**100.0**	**83.6**	**0.81**	**0.85**	**0.83**

that reject hallucinated scenarios before they enter downstream simulation, a critical requirement for regulated stress testing where implausible scenarios undermine stakeholder confidence.

Component contribution analysis. The ablation results reveal asymmetric contributions: removing validation gates increases invalid scenarios from 0% to 22.2%, confirming that unfiltered LLM outputs frequently violate economic constraints. Removing RAG grounding eliminates evidence links entirely (0% Evidence score) but maintains reasonable pass rates (80.2%), indicating that RAG primarily contributes *auditability* rather than *quality*—a crucial distinction for deployment in regulated environments where traceability is non-negotiable[36]. Removing structured schema degrades coherence to 75.9%, reflecting that free-form outputs lack the parametric precision required for downstream simulation. The interaction effects (not shown) reveal that RAG + gates together provide super-additive benefits: RAG surfaces relevant crisis precedents that make gate validation more effective by providing context for plausibility assessment.

Practical implications. The 5.4-second average generation time (vs. 18.2s for Reverse-ST) enables interactive exploration *of the scenario-design step*, where risk managers can iterate on stress narratives with rapid feedback. We stress that this figure covers only narrative-to-constraints generation; it is *not* the end-to-end latency. The downstream Monte Carlo simulation (MC=1000 replications per scenario over a multi-day horizon) dominates total cost—on the order of tens of seconds to a few minutes per scenario on our single-CPU setup—so "interactive" applies to scenario authoring, while full evaluation remains a batch operation. Generation time is therefore a small, fixed overhead relative to simulation, which is the practically relevant point: adding RAG-grounded generation does not materially change the end-to-end budget. The 96.2% traceability rate means that nearly every

generated constraint can be linked to supporting evidence (historical episodes, regulatory guidance, or economic principles), enabling audit-ready documentation that satisfies GAP-5 requirements. These results collectively confirm that the integrated narrative-to-constraints pipeline (Fig. 3) is necessary for simultaneously achieving quality, auditability, and efficiency, directly addressing GAP-1's scenario bottleneck.

Coverage analysis across the scenario space (Supplementary Table S1) confirms that MacroRisk-Agent achieves superior coverage (88 bins, 100% mechanism coverage) while maintaining plausibility.

Table 9 (Panel A) isolates the contribution of each pipeline component, revealing a modular architecture where components serve distinct functions.

Validation gates as quality filters. Removing validation gates increases invalid scenario rate from 0% to 22.2%, with pass rate actually *increasing* by 7.4 points. This paradox reveals the gates' function: they filter scenarios that would pass superficial checks but violate deeper economic constraints. Without gates, the system produces more "apparently plausible" but actually invalid scenarios—a dangerous failure mode for regulated applications where false confidence in flawed scenarios can lead to inadequate capital buffers.

RAG as auditability engine. Removing RAG grounding causes Audit score to drop from 96.2% to 44.3%, but pass rate decreases by only 12.4 points. This asymmetry confirms that RAG's primary contribution is *evidentiary grounding* rather than generation quality. In regulatory contexts where supervisors require traceable justification for stress assumptions[36], this auditability function is non-negotiable. The RAG retrieval surfaces relevant crisis precedents that anchor LLM outputs in documented mechanisms, enabling the "show your work" transparency that distinguishes acceptable from unacceptable stress testing.

Multi-turn refinement as error correction. The largest single-component impact comes from removing multi-turn refinement (ΔPass=−17.6, ΔPlaus=−16.2). This finding highlights that initial LLM outputs frequently contain subtle inconsistencies that iterative refinement resolves—for example, correcting magnitude mismatches between narrative severity and constraint parameters, or resolving temporal ordering violations. The refinement loop implements a form of self-consistency checking that substantially improves output quality without requiring external validation.

Design implication. The component contributions are largely orthogonal: RAG contributes auditability, gates contribute validity, schema contributes coherence, and refinement contributes consistency. This modularity implies that practitioners can selectively deploy components based on use-case requirements—for example, omitting RAG for internal exploratory analysis where traceability is less critical, while requiring full pipeline for regulatory submissions.

Table 9 (Panel B) demonstrates that MacroRisk-Agent maintains strong performance under distribution shift, with important implications for forward-looking stress testing.

Graceful degradation pattern. Across five distribution shift categories, the system retains an average 86.1% pass rate (vs. 92.6% baseline) and 92.7% traceability. The 5–9 percentage-point degradation is notably graceful: rather than catastrophic failure (which would indicate over-reliance on in-distribution patterns), performance declines smoothly as scenarios move further from training distribution. This robustness property is essential for stress testing, where the most valuable scenarios are precisely those that have *not* been observed historically.

Heterogeneous shift sensitivity. Performance varies across shift types in interpretable ways. Crypto-correlated scenarios show the largest degradation (83.4% pass rate, 77.6% evidence) because cryptocurrency dynamics are under-represented in the crisis library and lack established regulatory guidance. Climate-transition scenarios show smaller degradation (88.6% pass rate) because, despite being novel, they share structural similarities with commodity-shock scenarios that are well-documented. This pattern suggests that robustness depends on the availability of *analogical precedents*—scenarios that share mechanistic structure even if not surface-level similarity.

Traceability preservation. Even under distribution shift, traceability remains above 90%, indicating that the RAG retrieval successfully surfaces *some* relevant evidence even for novel risk configurations. This property is intended to support supervisory-style documentation (we make no claim of regulatory acceptance, which would require evaluation by regulators): a supervisor may tolerate moderate degradation in scenario quality for novel risks, but would be unlikely to accept scenarios that lack any evidentiary justification.

Implications for emerging risks. These proof-of-concept results suggest that, in principle, the prototype could support exploratory forward-looking analysis of emerging risks (climate transition, crypto contagion, AI-related operational risks) with appropriate confidence calibration, subject to validation on realistic data. Within the evaluated distribution shifts, novel risk categories produced validity rates 5–10 percentage points below the baseline condition; this observation motivates, but does not validate, a hypothesis that low-evidence scenarios would require additional expert review in any real application.

RQ2: systemic propagation with endogenous feedback

Experimental Setup. We evaluate propagation modeling on Suite A (16 severe scenarios, MC=1000 per scenario, 5 seeds). Baselines span supervisory approaches and academic systemic risk models: (i) *One-Pass* [BCBS'18]: single-round revaluation; (ii) *LiquidityAtRisk* [Cont+'20]; (iii) *IMF-SLST* [Oura'22]; (iv) *ContagionAcct* [Aldasoro+'22]; (v) *OverlapPort* [Poledna+'21]; (vi) *ScenFree* [Wiersema+'23].

Table 10 shows that, within the calibrated simulation, MacroRisk-Agent's endogenous feedback mechanisms produce larger systemic tail-risk estimates than the evaluated one-pass and single-channel configurations. This is an internal comparison of model configurations and does not establish that one-pass methods systematically underestimate real-world crisis severity.

Amplification magnitude in the calibrated setting. The endogenous-feedback configuration produces a 52% higher systemic-loss estimate than the one-pass configuration under the calibrated experimental setting. In absolute terms, simulated $SysES_{99}$ increases from 12.8% (One-Pass) to 19.4% (MacroRisk-Agent), a 6.6 percentage-point difference. Because the synthetic SysES measure has not been mapped to Basel III capital requirements, this difference should be interpreted as motivating questions for capital-planning analysis, not as a quantified capital consequence or evidence that any buffer is inadequate.

Market impact dynamics. The spread multiplier (1.59×) and depth contraction (35%) fall within the March 2020 calibration ranges, where Treasury bid–ask spreads widened and market depth contracted substantially[38]. Because March 2020 is used as a calibration anchor for γ, this agreement is an in-sample fit check rather than external validation of the liquidity module. The simulated forced-selling peak (7.2% of AUM) reflects the modeled fire-sale loop in which simultaneous liquidation depresses prices and triggers further de-risking.

Clustering and cascade dynamics. Throughout, we use the single term *cascade depth* for the number of propagation rounds until convergence (the "Cascade" column of Table 10), and use it consistently in place of the synonyms "cascade rounds" and "propagation rounds". The identification of 7 simultaneously constrained institutions and a cascade depth of 5.4 captures a phenomenon entirely absent from one-pass models (cluster=1, cascade depth=0): the *synchronization of distress*. This clustering emerges endogenously from similar constraint structures (leverage limits, margin requirements) combined with correlated exposures and common liquidity pools. The cascade depth of 5.4 indicates that amplification unfolds over multiple rounds—a dynamic that cannot be captured by single-shot revaluation and that is critical for understanding how localized shocks become systemic crises.

Comparison with academic baselines. MacroRisk-Agent produces larger simulated amplification than the evaluated LiquidityAtRisk[9], IMF-SLST[21], and ScenarioFree[19] implementations. The comparison is structurally asymmetric because the proposed simulator jointly models multiple feedback loops while these configurations intentionally isolate fewer channels. OverlapPort captures overlapping holdings but omits the liquidity loop; ContagionAcct captures accounting contagion but omits behavioral responses. The result therefore characterizes the effect of including all modeled loops, not a like-for-like superiority claim.

On the fairness of this comparison. This comparison is *structurally asymmetric*: each academic baseline (OverlapPort, ContagionAcct, LiquidityAtRisk) targets a single channel by design, whereas MacroRisk-Agent models four coupled loops, so a larger amplification is expected almost by construction. The 1.52× figure should therefore be read as a measure of *architectural completeness*—the consequence of jointly modeling channels that the single-channel methods deliberately omit—rather than as evidence that our method is intrinsically "better" at

any one channel. Two points make this interpretation precise. First, the RQ2 amplification is produced entirely by the deterministic multi-loop simulation engine; the LLM/RAG layer enters only at the scenario-generation stage (RQ1) and plays no role once Ω_s is fixed. Consequently, the relevant "full-channel, non-LLM" reference is exactly MacroRisk-Agent's own simulation core driven by fixed (non-generated) scenarios—which is what the loop-ablation study (Table 11, "All loops disabled (One-Pass)" through the full configuration) isolates: the move from 12.8% to 19.4% $SysES_{99}$ is attributable to the union of loops, not to agentic orchestration. Second, the contribution of the LLM/RAG layer is therefore *not* amplification magnitude but scenario coverage, evidence grounding, and mechanism attribution (RQ1 and Table 12A). Accordingly, the result should not be read as a like-for-like methodological win, a caveat we restate in the Limitations.

Temporal dynamics analysis showing how systemic risk accumulates over the stress horizon (T=5d, 10 d, 21 d) is provided in Supplementary Table S2, confirming non-linear risk accumulation with 62% increase from T=5d to T=21d.

Table 11 decomposes the contribution of each feedback mechanism to systemic amplification, providing simulation-level evidence for model design and hypotheses for future risk-management study.

Mechanism hierarchy. Collective selling emerges as the dominant driver, contributing 90.9% of excess loss (SysES drops by 6.0 points when disabled). This finding aligns with the theoretical prediction that coordinated deleveraging produces the most severe price impacts because it concentrates selling pressure in time and across correlated assets[9]. Endogenous liquidity dynamics contribute 54.5%, capturing the Brunnermeier-Pedersen spiral where market illiquidity and funding illiquidity reinforce each other. Margin procyclicality adds 34.8%, reflecting the volatility-haircut-liquidation feedback that regulators have increasingly sought to address through countercyclical buffer requirements.

Interaction effects and non-additivity. The individual contributions sum to 252.9% (90.9 + 54.5 + 34.8 + 18.2 + 30.3 + 24.2), far exceeding 100%, which indicates substantial *interaction effects*. When collective selling is disabled, endogenous liquidity has less material to amplify; when liquidity dynamics are disabled, margin calls have smaller price impacts. This non-additivity has a crucial implication: models that capture only a subset of loops will underestimate not only the missing loops' direct contribution but also their interaction with modeled loops. The full MacroRisk-Agent achieves 19.4% SysES versus the 12.8% of One-Pass, a 52% amplification that cannot be attributed to any single mechanism.

Overlapping holdings: small direct, large indirect effect. Overlapping holdings contribute only 18.2% directly, yet disabling this loop reduces margin events from 12.8 to 12.1 and forced selling from 7.2% to 6.5%. This pattern reveals the indirect role of common holdings: they do not directly cause losses but amplify the transmission of stress across institutions. When Institution A liquidates a commonly-held asset, Institution B incurs mark-to-market losses that may push it toward constraint boundaries, setting up future forced selling. This "common-holding contagion" channel has been identified as a key driver of systemic fragility in concentrated markets[9].

Simulation-level intervention hypotheses. Within the evaluated engine, the mechanism decomposition associates collective selling with de-synchronizing constraint boundaries, liquidity spirals with modeled backstop or circuit-breaker actions, and margin procyclicality with countercyclical haircut-floor actions. These associations are hypotheses generated inside the simulation, not recommendations for regulatory or market deployment; testing them would require realistic portfolios, implementation constraints, and observed intervention outcomes.

Table 12 (Panel A) evaluates mechanism attribution accuracy—the ability to correctly identify which feedback loop dominated each scenario's amplification.

Diagnostic capability within the simulation. MacroRisk-Agent achieves 85.4% Macro-F1, outperforming ContagionAcct (60.4%, $p < 0.001$) and the other evaluated baselines on the simulator-defined mechanism labels. The attribution links each scenario to the feedback loop recorded as dominant in the logs. It can therefore motivate candidate actions for counterfactual testing inside the same engine, but it does not establish which intervention would be appropriate or effective in a real market.

Discrimination quality. The 0.93 AUC-ROC measures ranking and discrimination, not probabilistic calibration. Within the evaluated scenarios, the score indicates that the system usually ranks a scenario whose logged dominant mechanism belongs to a given class above a scenario whose logged dominant mechanism does not. We therefore make no calibration claim from AUC-ROC; establishing probabilistic calibration would require a calibration-specific analysis such as a Brier score, calibration slope/intercept, ECE, or reliability diagram.

Baseline limitations. ContagionAcct and OverlapPort achieve only 55–60% F1 because they model single contagion channels and thus attribute all amplification to their modeled mechanism even when other loops dominate. ScenFree achieves lower accuracy (48.8%) because its scenario-free approach lacks the structured feedback-loop representation needed for attribution. These results demonstrate that mechanism attribution requires *explicit modeling* of multiple loops—a capability that distinguishes MacroRisk-Agent from both traditional and recent academic approaches.

Table 12 (Panel B) reports how MacroRisk-Agent responds to parameter perturbations, with sensitivity patterns that motivate testable hypotheses for risk monitoring and intervention timing.

Liquidity as the largest simulated amplifier. Low liquidity has the largest marginal impact (+32.0%) in the sensitivity sweep. When market depth is halved ($\ell \times 0.5$), SysES increases from 19.4% to 25.6%, and the amplification ratio rises from 1.52 to 1.82. This modeled sensitivity is consistent with the Brunnermeier–Pedersen liquidity-spiral mechanism, but it only motivates evaluation of liquidity indicators as possible early-warning variables; it does not establish an operational monitoring rule.

Volatility and correlation effects. High volatility ($\sigma \times 1.5$) increases SysES by 24.7% in the simulator, primarily through the margin-procyclicality channel. High correlation ($\rho \times 1.2$) has a smaller simulated effect (+12.4%) but increases cluster size from 7 to 8, indicating synchronization in the modeled agent population. This result suggests a hypothesis about correlation-induced clustering; it does not directly imply simultaneous failures or a regulatory response in real institutions.

Regime sensitivity. The crisis regime produces the most severe outcomes: SysES rises to 28.1% (+44.8%), amplification ratio reaches 1.96, and 12 institutions are simultaneously constrained. This configuration represents a "perfect storm" where all amplification channels are elevated simultaneously. Conversely, recovery conditions reduce risk by 23.7% (SysES falls to 14.8%), demonstrating that MacroRisk-Agent's risk estimates are not mechanically pessimistic but respond appropriately to improved conditions.

Implications for future stress-testing calibration. These sensitivity profiles show how scenario outputs change with assumed market conditions inside the prototype. They motivate future testing of condition-dependent calibration, but the reported marginal effects should not be applied to real-time market indicators or institutional stress tests without an externally validated mapping.

RQ3: simulation-defined controls and provenance-tracked explanations

Experimental setup. Suite C contains 50 severe scenarios with ranked intervention bundles. Each intervention is validated via counterfactual re-simulation under identical Ω_s. We evaluate: (i) intervention efficacy, (ii) decision quality at various top-K, (iii) auditability metrics, and (iv) cost-benefit tradeoffs. An important caveat applies to all RQ3 results: "success" and "improvement" are *simulation-defined*, i.e., measured as risk reduction inside the same simulation engine that generated the risk dynamics. This establishes internal consistency of the recommend–validate loop, but is *not* external validation of real-world intervention effectiveness; demonstrating the latter would require historical backtesting or expert evaluation of intervention plausibility, which we identify as future work.

Precise definition of "success". To avoid ambiguity we state the exact criterion. An intervention a for scenario s is counted as a *success* iff, under counterfactual re-simulation with identical constraints Ω_s, it (i) reduces the objective Φ (aggregate simulated tail loss, here SysES_{99}) by at least 15%; (ii) also reduces expected shortfall of the worst portfolio; (iii) satisfies all feasibility constraints (capital, leverage, liquidity limits); (iv) remains a net improvement *after* intervention costs; (v) introduces no metric-worsening side effect on the monitored risk

vector; and (vi) is stable in sign across the five seeds. Because every criterion is evaluated inside the simulator, the metric is more precisely named the *top-K simulated intervention-selection success under the predefined counterfactual objective*, and we use "success@*K*" only as shorthand for it. We benchmark against a transparent, non-learned policy heuristic (rank by largest exposure; Table 14) in addition to learning-based baselines, so the reported gains are not solely relative to system-specific methods.

Table 13 demonstrates that MacroRisk-Agent's counterfactual intervention framework delivers substantial risk reduction, with important implications for stress-testing governance and capital planning.

Intervention effectiveness hierarchy. Single-mechanism interventions show heterogeneous effectiveness: leverage tightening (−13.6% SysES) outperforms liquidity buffers (−12.1%) and HY de-risking (−10.7%), while margin add-ons (−6.8%) and index protection (−8.7%) have smaller impacts. This hierarchy reflects the underlying amplification dynamics: leverage constraints directly truncate the fire-sale spiral by limiting position sizes, while liquidity buffers provide defensive capacity without reducing exposure. The practical implication is that "one-size-fits-all" intervention guidance is suboptimal; the most effective action depends on which feedback loops dominate the scenario.

Synergistic effects of combined interventions. The optimal agent-recommended bundle achieves 33.0% SysES reduction, which falls short of the naive additive sum of the best three single interventions (13.6 + 12.1 + 10.7 = 36.4% if additive, versus an actual combined effect of 33.0%). This *sub-additivity* reflects that interventions addressing the same feedback loop have diminishing marginal returns, while interventions targeting different loops are complementary. The agent learns to combine liquidity buffers (truncating liquidity spirals) with de-risking (reducing fire-sale volume) and protection (providing downside insurance)—a portfolio of interventions that addresses multiple amplification channels simultaneously.

Worst-portfolio vs. systemic trade-offs. The optimal bundle reduces worst-portfolio ES by 37.7% versus SysES reduction of 33.0%, indicating that the intervention particularly benefits the most vulnerable portfolios. This distributional effect is desirable from both microprudential (protecting individual institutions) and macroprudential (preventing concentrated failures) perspectives. In contrast, macro hedges + de-risking achieve only −15.0% SysES and −18.0% worst-portfolio ES, revealing that hedging-focused strategies provide less systemic benefit than buffer-focused strategies.

Forced-sale reduction as leading indicator. The 48.1% reduction in forced selling (from 7.9% to 4.1% of AUM) is particularly significant because forced selling is the proximate cause of fire-sale spirals. By reducing forced selling, the optimal bundle truncates the amplification mechanism at its source rather than merely absorbing losses after they occur. This "upstream" intervention logic—targeting the *drivers* of systemic risk rather than the *symptoms*—is enabled by MacroRisk-Agent's mechanism-aware simulation that tracks forced selling as an explicit state variable.

Implications for capital planning. These results inform the cost-benefit analysis of capital and liquidity buffers. A 30% increase in liquidity buffers produces 12.1% SysES reduction (Table 13), implying a "risk-reduction multiplier" of approximately 0.4×. Risk managers can use these multipliers to calibrate buffer levels to target residual risk levels, rather than relying on fixed regulatory minimums that may be insufficient in severe scenarios.

Table 14 evaluates simulation-defined decision quality at various top-*K* recommendation depths, addressing the within-environment question: *How many candidate interventions must be re-simulated before the candidate set contains an action that meets the six simulator-level success criteria?*

Top-1 decision quality. MacroRisk-Agent achieves 68.6% success@1 (vs. 61.4% for RL-policy, $p < 0.01$), meaning the top-ranked intervention achieves a simulation-defined risk reduction of ≥15% in over two-thirds of scenarios. This 7.2 percentage-point improvement has substantial practical value: in crisis situations where decision time is limited, the ability to identify effective interventions on the first recommendation reduces the iteration cost that can delay response. The 21.8% average improvement at K=1 (vs. 13.8% for heuristic) indicates that MacroRisk-Agent not only identifies *effective* interventions but identifies *high-impact* ones.

Diminishing returns at higher K. Success rates increase from 68.6% (K=1) to 86.8% (K=3) to 90.4% (K=5)—diminishing marginal gains that reflect the structure of the intervention space. By K=5, most scenarios

have at least one effective intervention in the recommendation set, but the residual 9.6% failure rate represents "hard" scenarios where no intervention from the action space achieves ≥15% reduction. These may be scenarios with multiple simultaneous tail shocks that overwhelm any feasible intervention, or scenarios where the stress narrative specifically targets intervention-resistant configurations.

Comparison with baseline strategies. Random selection achieves 17.4% success@1 under the simulator-defined criterion. Heuristic ranking (by exposure size) achieves 44.6%, while gradient-based and RL-policy approaches achieve 57.2% and 61.4%, respectively. MacroRisk-Agent provides the best performance among these evaluated baselines, with a 7.2-point difference from RL-policy. This comparison concerns selection inside the shared simulator and does not demonstrate real-world intervention effectiveness.

Top-K coverage within the evaluated simulation. Evaluating the top three candidates captured at least one simulator-defined successful intervention in 86.8% of the evaluated scenarios, while the top ten captured one in 95.4%. The additional simulated coverage beyond ten candidates was 2.2 percentage points. These figures describe only the fixed action library and synthetic scenarios used here; they do not justify limiting practitioner attention, setting an operational search depth, or acting on the ranked candidates without external review.

Table 15 evaluates prediction calibration (Panel A) and auditability (Panel B) within the simulated environment; neither panel by itself establishes trustworthy real-world decision support.

Within-simulator prediction agreement. MacroRisk-Agent achieves 5.6% calibration error at top-1 (vs. 7.4% for RL-policy, $p < 0.01$). Thus, within this simulator, a predicted 25% objective reduction is associated on average with a realized reduction within 5.6 percentage points. This is agreement between prediction and re-simulation under the same model, not calibration to actual market or capital outcomes.

Correlation and ranking quality. The $R^2 = 0.82$ indicates that 82% of the variance in simulator-realized improvements is explained by predicted improvements. The Spearman correlation of 0.88 measures ranking agreement among candidates within the evaluated action library and simulator; it should not be interpreted as evidence that implementing the top-ranked intervention would be reliable in practice.

Error distribution analysis. The error bounds (±1.3%) indicate limited variation across the evaluated synthetic scenarios. They do not establish invariance under new scenario families, real portfolios, or distribution shift, and the same intervals should not be applied operationally without external validation.

Comparison with alternatives. Heuristic approaches show 12.8% calibration error with $R^2 = 0.40$, indicating that simple rules provide poor prediction of intervention effectiveness. Even sophisticated approaches like gradient-based optimization (8.9% error, $R^2 = 0.62$) and RL-policy (7.4% error, $R^2 = 0.71$) substantially underperform MacroRisk-Agent. The improvement stems from the counterfactual validation loop (Fig. 5), which explicitly simulates intervention effects under the same scenario constraints rather than extrapolating from historical correlations.

Provenance and traceability. MacroRisk-Agent achieves 97.6% provenance completeness and 89.2% traceable claims, meaning that most system outputs resolve to specific inputs (scenario constraints, crisis precedents, regulatory guidance, or simulation parameters). This supports a "rewind-and-explain" inspection of which assumptions, evidence records, parameters, and computations produced a recommendation. The mechanism is aligned with the documentation principle that stress-test assumptions should be transparent and well documented[36], but the measured traceability rate does not establish regulatory acceptance or explanatory correctness.

Explanation–log consistency. The 83.6% faithfulness score measures machine-checkable agreement between mechanism statements in the LLM-generated explanation and the dominant mechanisms recorded in the simulation logs. The corresponding F1 score is 0.83 (precision 0.81; recall 0.85). These values support an internal log-consistency claim only: an explanation may match an economically misspecified simulator, and the metric neither verifies factual or causal correctness nor guarantees that an explanation is trustworthy.

Scoring protocol for the auditability metrics. For transparency we state exactly how Panel B was computed. *Provenance completeness* and *evidence traceability* are scored *automatically by rule-based matching against the simulation and retrieval logs*: a claim is "traceable" if its referenced scenario id, evidence id, metric, or seed resolves to an existing record in the reproducibility object $\mathcal{R}$. *Schema* is a hard JSON-validation check.

Faithfulness is scored automatically by aligning each sentence of the LLM explanation that asserts a mechanism (e.g., "margin spiral dominated") against the logged dominant-loop attribution for that scenario; agreement counts toward precision/recall. No human expert annotators were used, so we do *not* report inter-annotator agreement; the metrics should be read as *machine-checkable consistency* between generated text and internal logs, not as human-judged correctness. The main failure mode of the automatic faithfulness scorer is paraphrase mismatch (an explanation describing the right mechanism in wording the matcher does not recognise), which biases the 83.6% figure *downward*; conversely, it cannot detect an explanation that is internally consistent with the logs but economically wrong. Concretely, a claim counts as *traceable* only if it resolves to a *direct source reference* (a retrieved evidence id) or a *deterministic simulation-log record* (scenario id, metric, seed) in $\mathcal{R}$; we deliberately do *not* accept looser *semantic entailment* as traceability, so a link certifies provenance, not explanatory correctness. Human-expert faithfulness evaluation is left to future work. Worked examples of a faithful and an unfaithful explanation are given in cases (d) of the Failure-Case Analysis.

Ablation analysis of auditability components. The ablation results show how each component contributes to the defined auditability metrics. Free-text-only outputs (no structured linking) achieve 21.4% traceable claims despite 100% provenance; No-RAG systems achieve 0% evidence score; and Partial RAG (top-3 only) achieves 71.6% evidence score. These comparisons identify design trade-offs inside the prototype, not requirements for regulated deployment.

Auditability implications. The auditability metrics directly address GAP-5's requirement for "actionable and auditable" stress testing. Supervisory examinations increasingly focus on model governance and documentation quality; stress tests that cannot explain their assumptions face elevated model risk scrutiny. MacroRisk-Agent's 89%+ traceability and 83%+ faithfulness are intended to support regulatory-style documentation and internal model-governance review; we do not claim regulatory acceptance, which would require evaluation by supervisors on real portfolios.

Cost-benefit analysis of individual and combined interventions is provided in Supplementary Table S3.

Failure-case analysis and a gate-independent manual author spot-check

Reporting only aggregate success rates can obscure how and where the system fails. We therefore catalogue four representative failure modes observed in our logs and report a small gate-independent manual author spot-check of the "0% schema-defined hallucination under the specified validation criteria" claim. The assessment is independent of the automated gates, but it is not an external or independent-expert evaluation.

(a) Invalid scenarios correctly rejected by the gates. A frequent rejection is a draft that sets a $+260$ bps high-yield spread shock while leaving the liquidity depth multiplier at 0.95, violating consistency rule C2 (wide credit stress should coincide with depth deterioration). The Plausibility gate also routinely rejects single-day equity moves beyond the historical 99th percentile (e.g., a -42% one-day drop) that the LLM occasionally proposes when prompted for "extreme" stress. These are *desirable* rejections: the gates are doing their job.

(b) Misleading scenarios that nonetheless passed the gates. More concerning are scenarios that are schema-valid yet economically questionable. In one case the system generated a simultaneous oil spike and equity rally that each lay within historical bounds and violated no pairwise rule, but whose *joint* configuration is implausible; our pairwise consistency gates (C1–C4) do not test all higher-order joint dependencies, so it passed. This is exactly the gap between "schema-defined" validity and economic truthfulness emphasized in Definition 4 and the Limitations.

(c) Failed interventions. The 9.6% of scenarios with no successful intervention at $K=5$ (Table 14) are typically "multi-binding" crises in which leverage, margin, and funding constraints bind simultaneously, so any single-lever action is offset by another tightening loop—an honest negative result that the simulation-defined metric surfaces rather than hides.

(d) Unfaithful explanations. The 16.4% of explanations not counted as faithful split into two groups: benign paraphrase mismatches (the explanation names the correct dominant loop in wording the automatic matcher

Table 16 Summary: MacroRisk-Agent vs. Best Evaluated Baseline Improvements. All p-values are from paired two-tailed *t*-tests on seed-averaged base-scenario differences after Shapiro–Wilk screening and Benjamini–Hochberg adjustment.

RQ	Key Metric	Improvement	*p*-value
RQ1	Pass@Gate (%)	+14.2 pts	<0.001
RQ1	Plausible@Strict (%)	+16.5 pts	<0.001
RQ1	Traceability (%)	+34.5 pts	<0.001
RQ2	$SysES_{99}$ (%)	+1.8 pts	<0.001
RQ2	Amplification ratio	+0.14	<0.01
RQ2	Mechanism F1 (%)	+25.0 pts	<0.001
RQ3	Success@1 (%)	+7.2 pts	<0.01
RQ3	CalibError (%)	−1.8 pts	<0.01
RQ3	Faithfulness (%)	+5.8 pts	<0.05

misses) and genuine errors in which the narrative over-attributes losses to the most salient loop (e.g., crediting "fire sales" when the logs show margin procyclicality dominated). Only the latter is a true faithfulness failure; we cannot currently separate the two automatically, which is why we frame faithfulness as machine-checkable consistency rather than correctness.

Gate-independent manual author spot-check. To partially address the self-referential nature of the hallucination metric, the authors manually reviewed a random sample of 30 *gate-accepted* Suite B scenarios against the retrieved evidence and basic economic sense. All 30 were schema- and rule-consistent (confirming the gates), but 4 (13%) were judged "economically weak" (a defensible-but-unlikely joint configuration of the type in case (b)). We therefore report "0% schema-defined hallucination under the specified validation criteria" alongside an estimated ∼13% rate of author-flagged economic weakness among accepted scenarios. This is a small author assessment independent of the automated gates, not independent expert validation, and the headline figure must be read as filter consistency rather than verified correctness.

Table 16 consolidates the main experimental findings across the three research questions. For each key metric, it reports the absolute improvement of MacroRisk-Agent over the strongest competing baseline (e.g., FinAgent-RAG for RQ1, OverlapPort for RQ2, RL-policy for RQ3). For every comparison in this summary table, the Shapiro–Wilk test on the paired base-scenario differences did not reject normality at $\alpha = 0.05$; accordingly, every reported p-value uses a paired two-tailed *t*-test after averaging the five nested seeds. The Wilcoxon fallback specified in the statistical protocol was not invoked for Table 16. P-values are Benjamini–Hochberg adjusted within each RQ metric family. **RQ1** metrics cover scenario quality (Pass@Gate, Plausible@Strict) and auditability (Traceability). **RQ2** metrics capture systemic tail risk ($SysES_{99}$), amplification magnitude (Amp), and mechanism identification accuracy (Mechanism F1). **RQ3** metrics assess decision support (Success@1), prediction calibration (CalibError), and explanation faithfulness (Faithfulness). All reported differences are statistically significant within the evaluated synthetic scenarios ($p < 0.05$ or better); they do not establish robustness to real-world portfolios, interventions, or distribution shifts outside the tested conditions.

Conclusion

This paper introduced *MacroRisk-Agent*, a multi-agent LLM architecture designed to constrain selected forms of structured-generation failure through deterministic validation gates for auditable structured decision support, instantiated in the domain of financial stress testing. The design addresses five technical gaps in deploying LLM-agent systems for high-stakes applications: constrained generation, semantic-to-numeric coupling, feedback modeling, multi-level aggregation, and auditability. MacroRisk-Agent integrates three core capabilities: (i) a RAG-grounded, schema-constrained generation pipeline with deterministic validation gates that achieves 92.6% validity with 0% schema-defined hallucination under the specified validation criteria (Definition 4); (ii) a constraint-coupled multi-agent simulation that produces a 52% higher systemic-loss estimate than the one-pass configuration under the calibrated experimental setting, with 85.4% mechanism attribution accuracy; and (iii) a

counterfactual re-simulation framework achieving 68.6% simulation-defined top-1 decision success rate with 83.6% machine-checkable explanation–log consistency. We report these as proof-of-concept findings obtained on synthetic portfolios and curated scenario libraries; they characterise the architecture's behavior within our evaluation environment rather than established performance in real institutional stress testing. Framed conservatively, the study provides controlled experimental evidence that the proposed architecture can improve defined validity, provenance, and counterfactual-evaluation metrics in a simulated stress-testing environment. The framework has *not* been validated on confidential institutional portfolios, regulator-approved models, real trading behavior, or actual intervention outcomes, and we make no claim of regulatory validation, operational readiness, or real-world supervisory suitability. The COVID-2020 diagnostic is an in-sample calibration-fit check because the price-impact coefficient uses that episode as an anchor, and the reported amplification and cascade-depth reference values are model-derived episode replays rather than directly observable market quantities.

The key architectural insight is the *separation of semantic and numerical responsibilities*: LLMs handle evidence retrieval, narrative synthesis, and explanation generation, while deterministic engines execute simulation, pricing, and metric computation. This separation, enforced through a structured schema interface and multi-stage validation gates, provides a principled approach to schema-level hallucination control that we hypothesize can generalize beyond the financial domain to other settings where LLM outputs must satisfy hard constraints and maintain provenance; testing this generalization is left to future work.

Limitations. MacroRisk-Agent's quantitative fidelity depends on the calibration of factor dynamics, liquidity response functions, and behavioral constraint rules. While the architecture captures key amplification channels, certain market microstructure details, institution-specific execution constraints, and full counterparty network dynamics may require richer data and more granular modeling. Scenario generation quality depends on the breadth and curation of the crisis/event library available to the retrieval layer. We expand below on the limitations that most directly bound our claims.

(1) Synthetic evaluation and external validity. All results are obtained on 10 synthetic portfolios drawn from a 6-ETF universe, a 150-entry crisis library, and a single simulator. Real institutional portfolios have far richer asset heterogeneity, OTC exposures, and idiosyncratic constraints. We therefore frame all findings as *proof-of-concept* and make no claim of deployment readiness; validation on real (anonymized) institutional data is the most important next step.

(2) Self-referential hallucination metric. The "0% schema-defined hallucination under the specified validation criteria" figure follows by construction from Definition 4: accepted scenarios cannot fail the gates that define them. The metric measures filter consistency, not independently verified correctness; our small manual spot-check (Failure-Case Analysis) already flags ∼13% of accepted scenarios as economically weak. Independent expert review or comparison against ground-truth crisis parameters is needed for a stronger claim.

(3) Baseline asymmetry. The RQ2 52% amplification reflects *architectural completeness*—modeling four coupled loops versus single-channel baselines—rather than a like-for-like methodological win. The amplification is generated by the deterministic engine, not the LLM/RAG layer; the loop-ablation study is the relevant fair comparison.

(4) Proprietary-LLM dependence. The backbone is GPT-4o; we report <3% degradation with GPT-4o-mini but only preliminary open-source results (Llama-3-8B) in the repository. Regulated environments that prohibit proprietary APIs (data residency, confidentiality) would require systematic quantification of gate robustness and schema compliance under open-source models, which we have not completed.

(5) Fixed- Ω_s *counterfactual assumption.* Interventions are re-simulated under *identical* scenario constraints Ω_s. Real interventions (e.g., leverage tightening) alter the dynamics themselves—de-risking changes agent composition and market microstructure—so holding Ω_s fixed may systematically *overstate* intervention efficacy. Endogenizing Ω_s under intervention is left to future work; until then, intervention "success" is simulation-defined.

(6) Calibration sensitivity of headline numbers. The 52% amplification depends on liquidity-impact, margin, and behavioral parameters. The sensitivity analysis (Table 12B) shows the amplification ratio remains > 1 and the qualitative ranking is preserved across a wide sweep (Amp ranges 1.26–1.96), so the *direction* is robust; however,

the precise magnitude is a function of selected liquidity-impact parameters and should not be treated as a calibrated point estimate. Because γ uses March 2020 as a calibration anchor, the COVID column in Table 5 is not held out and provides no out-of-sample evidence.

(7) Coverage saturation. The 150-entry library is small relative to the 500-scenario RQ1 evaluation. Consistent with a saturation effect, generation quality degrades as scenarios move away from library coverage (Table 9B): crypto-correlated scenarios, the least represented, show the largest drop (Pass 83.4%, Evidence 77.6%). Forward-looking use of under-represented risk families should expect lower validity and be paired with expert review.

(8) Compute scalability and statistical precision. Experiments run on a single CPU (Apple M2) with 10 portfolios; end-to-end cost is dominated by Monte Carlo simulation, which scales roughly linearly in (portfolios × horizon × MC paths). Institutional-scale portfolio sets and longer horizons would require parallelization and were not benchmarked. Statistically, the independent unit is the scenario (not the seed), and bootstrap intervals are clustered by scenario family; however, five seeds per scenario remain too few to estimate seed-level variance components precisely. Here a scenario-family cluster is one base scenario together with its five nested realizations; Suites A–C contain 16, 500, and 50 such clusters, respectively.

Future work. Promising directions include: (i) extending the validation-gate framework to other high-stakes domains (clinical decision support, infrastructure planning) where constrained generation and auditability are critical; (ii) integrating real-time data feeds for continuous scenario refresh and adaptive stress testing; (iii) developing standardized benchmarks for evaluating hallucination control in structured LLM outputs; and (iv) exploring fine-tuned open-source LLMs to reduce dependence on proprietary APIs while maintaining generation quality. Most importantly, the proof-of-concept status calls for (v) validation on real (anonymized) institutional portfolios; (vi) an independent, expert-based hallucination and faithfulness assessment; (vii) endogenizing the scenario constraints Ω_s under intervention so that counterfactuals reflect altered dynamics; and (viii) historical backtesting of recommended interventions to move from simulation-defined to real-world effectiveness.

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Data availability All market factor data used in this study are publicly available from the FRED database (https://fred.stlouisfed.org) and Yahoo Finance API. The crisis event library (150 entries), synthetic portfolio configurations, prompt templates, and simulation code are available at the project repository: https://github.com/YanLiuSIIT/macrorisk-agent. LLM-generated outputs for all experiments are cached and included in the repository to ensure reproducibility without API access. A subset of experiments can be reproduced using open-source LLM alternatives (e.g., Llama-3-8B-Instruct) as specified in the repository README.

Declarations

Competing interests The authors declare no competing interests.

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